FigSpec FS60-C Hyperspectral Camera: Operational, Data Processing and Troubleshooting Guide

Prepared

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This guide is applicable to the FigSpec FS60-C hyperspectral camera with serial number SN/FSW255030.

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1. Product Parameters

Lighting Mode	Passive lighting (without light source)
Spectroscopic Method	Transmission grating
Spectral Range	400-1000nm
Spectral Band	300(4), 600(2), 1200(1) (2nm, 1nm, 0.5nm)
Spectral Resolution (FWHM)	2.5nm
Slit Width	25um
Transmission Efficiency	>60%
Stray Light	<0.5%
Number of Spatial Pixels	480, 960, 1920 (software configurable)
Pixel Size	5.86um
Imaging Speed	Full band 128Hz,after ROI can achieve 3300Hz
Probe	CMOS
SNR(Peak)	600/1
Camera Output	USB3.0 or Gigabit network
Camera Interface	C-Mount
Attachment	USB3.0 transmission line or Gigabit network transmission line
ROI	Multiple regions
Embedded Data Acquisition Processing Storage Unit	N5105 processor 1T/512G SSD storage
Serial Number	FSW255030

2. Initialization Settings

2.1. Mounting the Camera onto the Drone

As the cameras used in this study, including the FigSpec FS60-C, DJI Zenmuse L2, and DJI H30T, must be operated individually rather than simultaneously, the use of a Single Gimbal Connector is recommended. This configuration provides better centralised weight distribution and balance during flight while also reducing the overall weight of the mounting assembly.



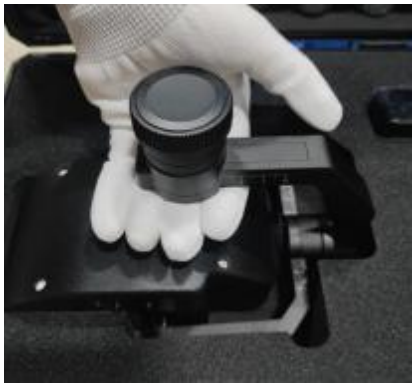
(a) Dual Gimbal Connector



(b) Single Gimbal Connector

Grasp the top of the gimbal, support it with your hand, and remove the camera, as shown in the diagram.



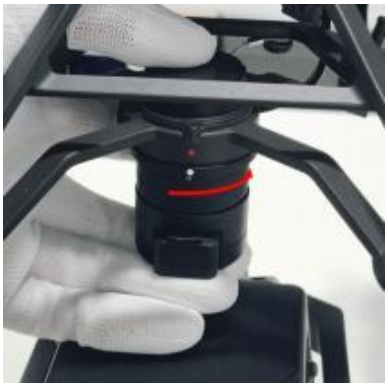


Remove the lens cap.



Rotate the gimbal protector counterclockwise and remove it. With the white dot facing towards you, attach it to the gimbal bracket of the drone. Rotate the gimbal until the red dot on the gimbal aligns with the red dot on the drone's gimbal bracket.





Camera Power On

1. Turn on the drone remote controller and power on the drone.

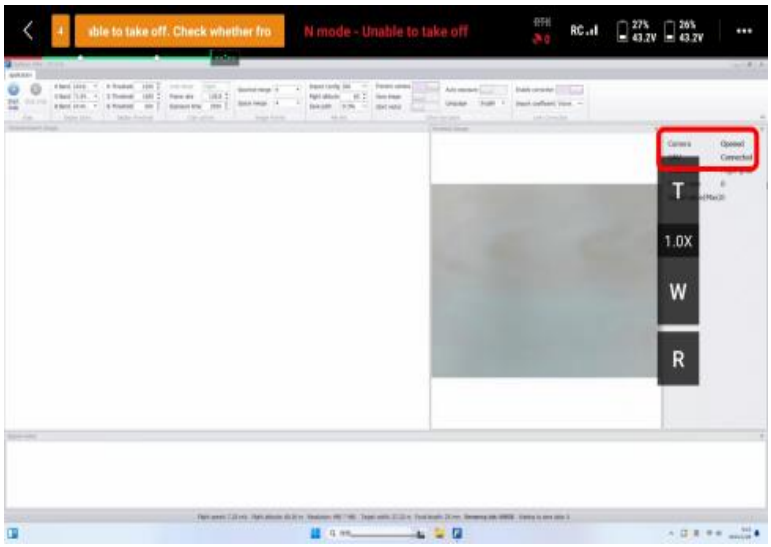
2. Wait for the camera self-check to complete and return to the center position. Press the camera power switch within 3 seconds.



3. Wait for the camera to power on. Once the camera is powered on, a small window named Figspec will appear in the bottom right corner of the remote controller screen.

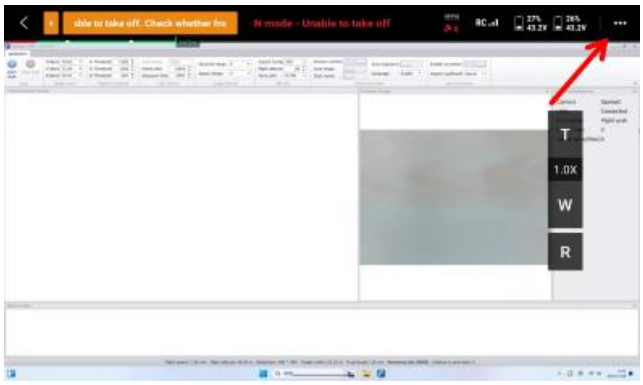


After the camera is powered on, the software will automatically start up. Once the software is launched, in the right-hand status bar, both the camera and the drone will be displayed as connected, indicating that the camera has been powered on correctly.

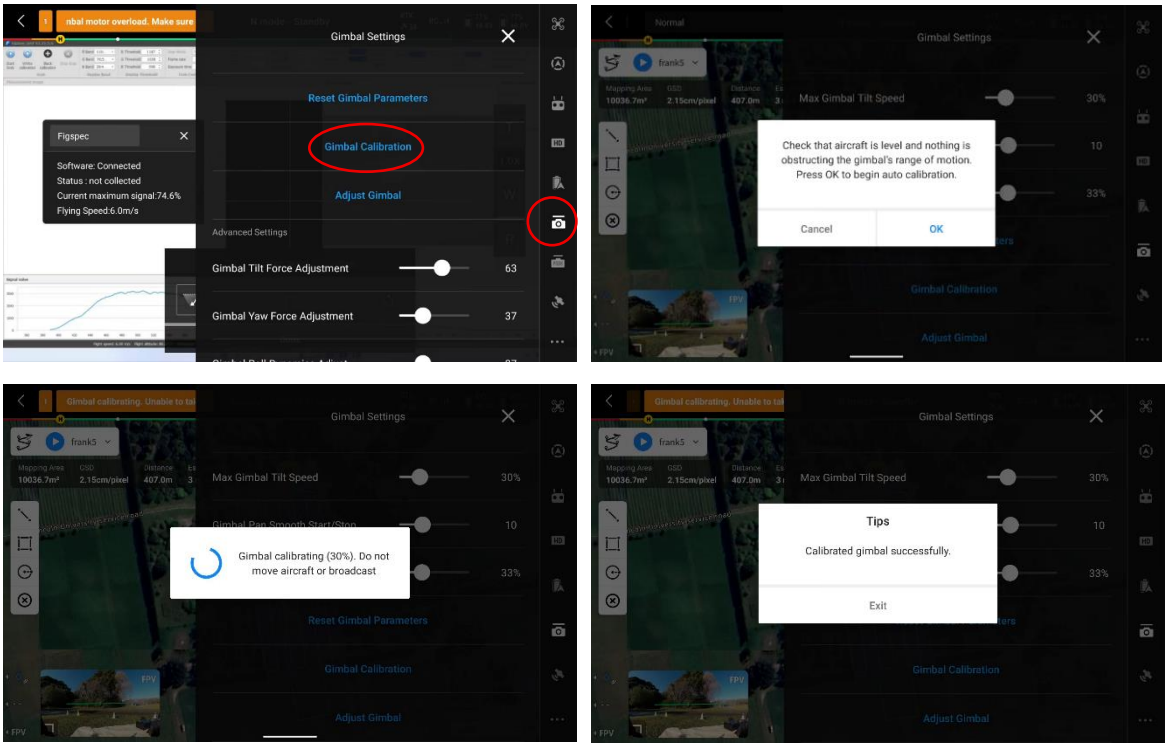


2.2. Gimbal Parameter Auto-tuning

Click on the three dots in the top right corner of the interface to access the settings menu.



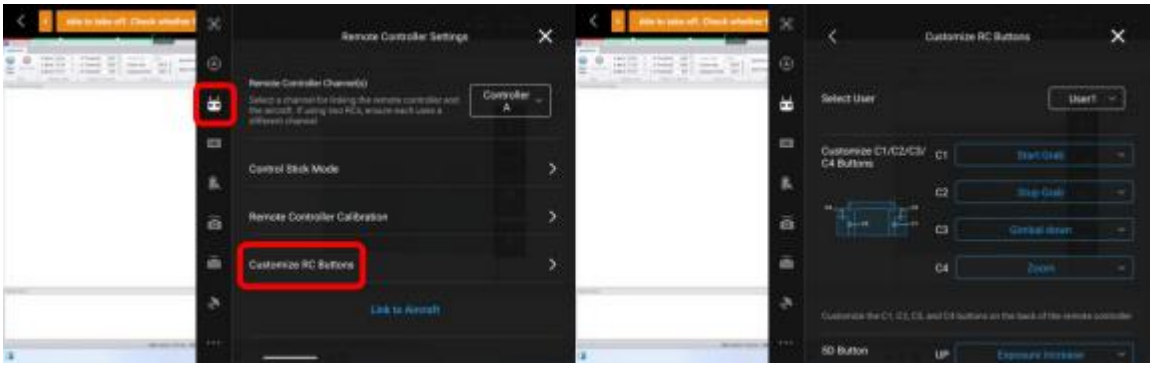
Click on the gimbal icon, then swipe up on the screen until reaching the bottom, and tap on "Gimbal Parameter Auto-adjust". The gimbal will then auto-adjust itself. Wait for the Auto-tuning to complete. Under normal circumstances, gimbal parameter calibration is not required.



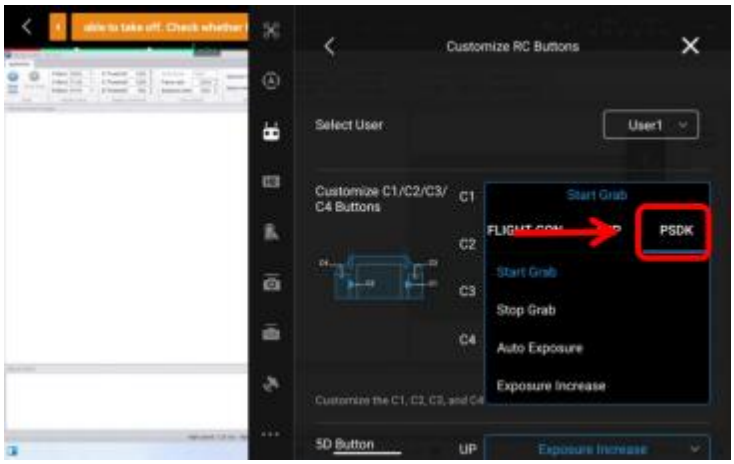
2.3. Shortcut Key Settings

Note: Shortcut key settings need to be done after connecting the hyperspectral camera.

1. Click on the three dots in the top right corner of the remote controller interface. Then, select "Remote Controller Icon" and proceed to "Customize RC Buttons" .

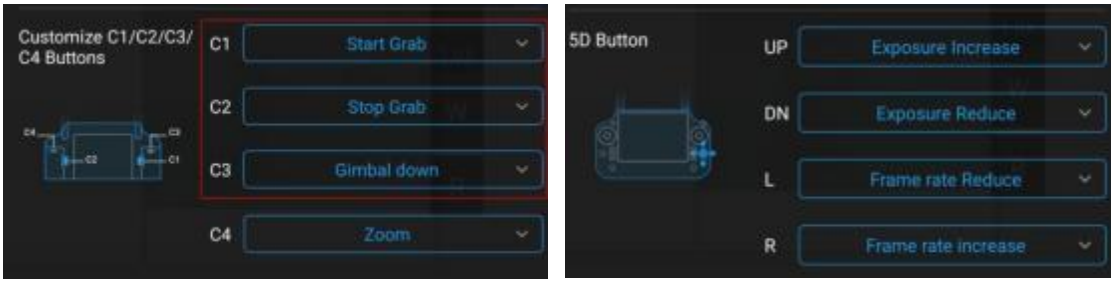


2. Set the corresponding buttons to your preferred shortcuts. Click on the dropdown menu, swipe left or right at the top to select the functional area, find “PSDK”, and then choose the corresponding function. For example, refer to “Shortcut Key”.



Configure other shortcut functions for instrument settings on different buttons.

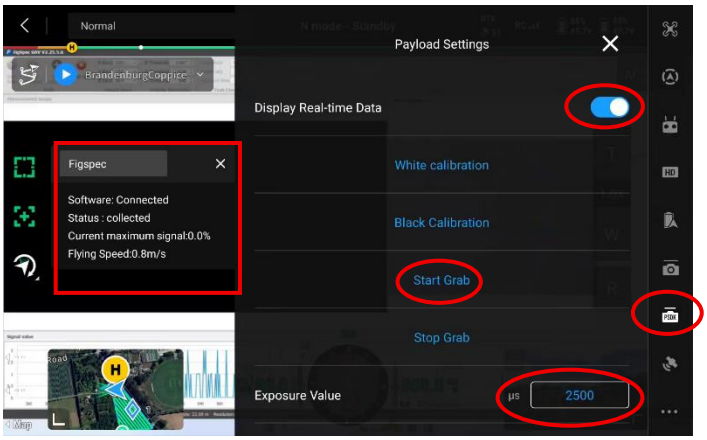
3. Shortcut Key Settings Template.



2.4. Adjust Exposure Value

The camera is an older version (serial number is FSW255030), and the previous user manual does not fully match the hardware. Some of the functions described in the manual are not available on our system. The white calibration button is not functional on our camera. There is no automatic exposure function on this version. After experimenting for a while, I finally figured out a complete method for adjusting exposure values and white calibrating with a white board (80% reflection) for this camera.

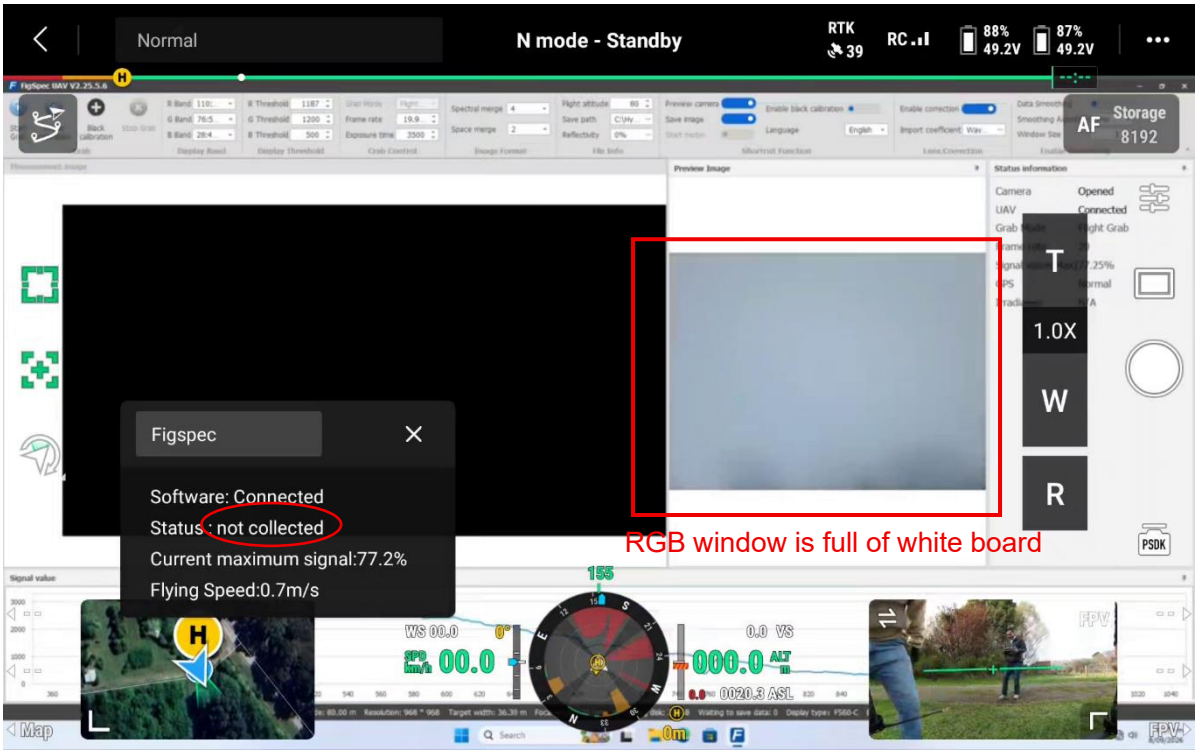
1. Open the Real-Time Data window, then click Start Grab. The real-time data window should appear as shown below, with both Software and Status displaying “Connected” and “Collected”, and the Current Maximum Signal showing dynamically changing values.。



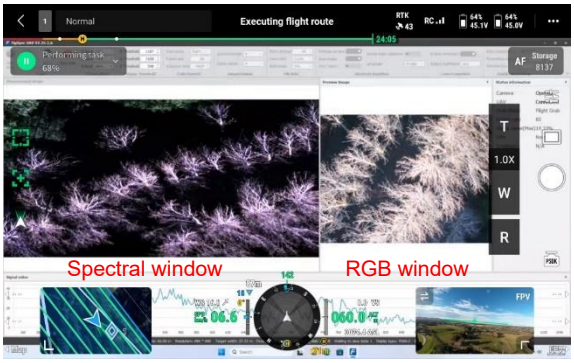
2. Place the white board on the ground, ensuring that it is fully illuminated by sunlight and that no shadows fall on its surface. On cloudy days, this is less of a concern. Then, hold the drone up with the camera pointing directly downward at the white board. Adjust the distance between the drone and the board until the white board completely fills the RGB window. At this point, the Current Maximum Signal value should fluctuate dynamically. If the value is below 80%, click Stop Grab and confirm that Status displays “Not Collected”. Then, increase the Exposure Value. Conversely, if the Current Maximum Signal exceeds 80%, decrease the Exposure Value. After each adjustment, click Start Grab again and monitor the Current Maximum Signal. Repeat this process until the value is close to 80%. Once the desired value is achieved, click Stop Grab and ensure that Status displays “Not Collected”. The Exposure Value adjustment is now complete.



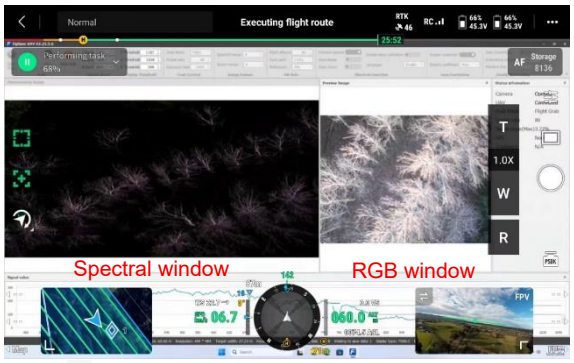



Normal
N mode - Standby
RTK
39
RC
88%
49.2V
87%
49.2V
Storage
8192
AF
Figspec
Software: Connected
Status : not collected
Current maximum signal:77.2%
Flying Speed:0.7m/s
T
1.0X
W
R
PSDK
RGB window is full of white board
Map
FPV

After the Exposure Value has been adjusted, the live camera view displayed on the remote controller during the flight mission should appear as shown in the figure below.



(a) Normal exposure value



(b) The exposure is a bit low

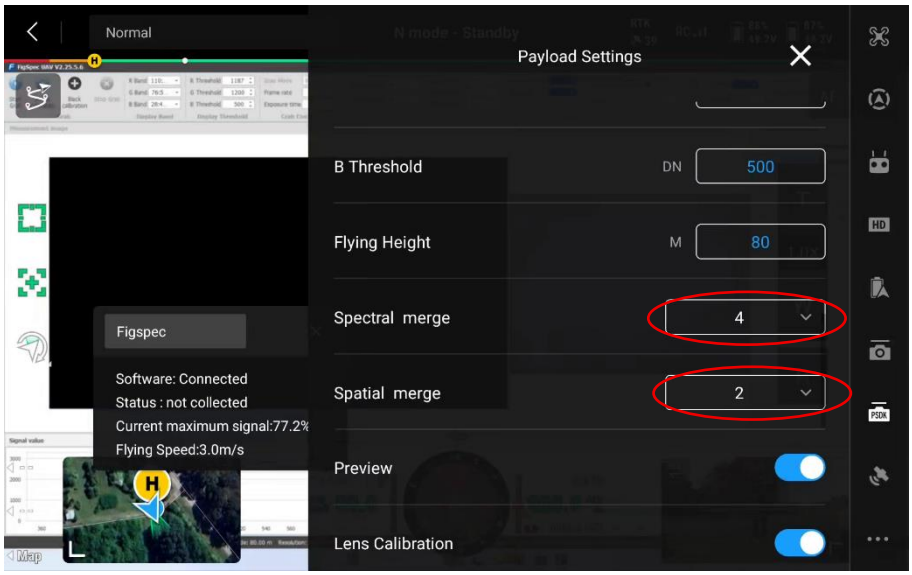
3. Once the exposure adjustment is complete, place the white reference panel within the flight mission area, ensuring that it is clearly visible to the camera during flight. Do not place the panel beneath trees or in locations where it may be obstructed. Avoid shaded areas and position the panel in an open area under direct sunlight.



2.5. Set Spacial Merge and Spectral Merge

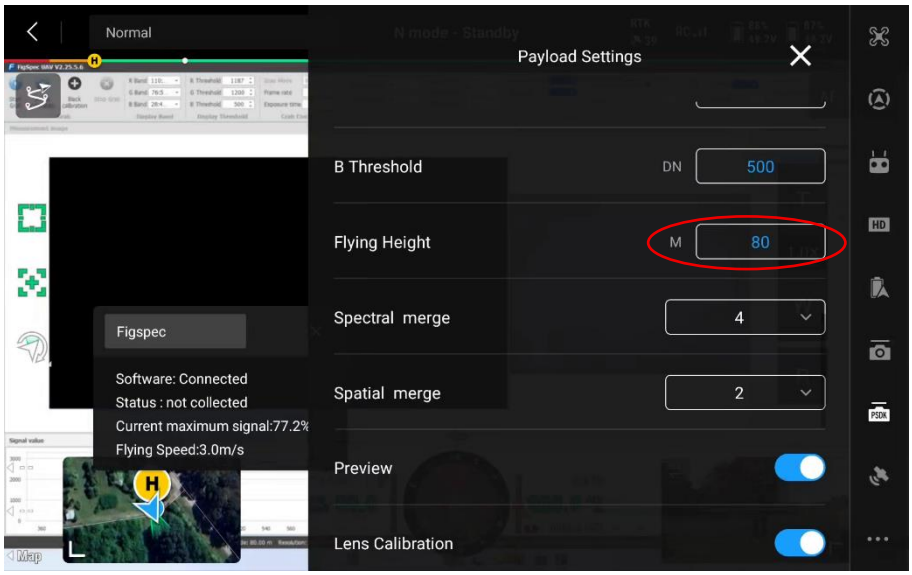
1. The Spectral Merge setting provides three options: 1, 2, and 4, corresponding to spectral sampling intervals of 0.5 nm, 1 nm, and 2 nm, respectively. Selecting 1 may reduce the effective signal sensitivity and result in a lower signal-to-noise ratio (SNR). Unless particularly fine spectral analysis is required, it is recommended to select 2 or 4. Option 1 is generally not recommended for routine data acquisition.

2. The Spatial Merge setting also provides three options: 1, 2, and 4. Selecting 1 provides higher spatial resolution, expressed in cm/pixel, but may result in a lower signal-to-noise ratio (SNR), larger data volumes, and the need for a longer exposure time. Therefore, Option 1 is generally not recommended for routine data acquisition. Instead, Options 2 or 4 are recommended, as they provide a better balance between spatial resolution, signal quality, data volume, and exposure requirements.



2.6. Set Flying Height

Regarding flying height, the manufacturer recommends an operating height of 80–120 m, as the camera's fixed-focus lens is factory-calibrated to provide optimal image sharpness and clarity within this range. The camera can still be operated outside this recommended altitude range; however, image sharpness may decrease due to the fixed-focus lens characteristics. Another important consideration is airspace approval. In general, higher flying height may make obtaining the necessary airspace authorisation more difficult. Therefore, flying height should be selected by balancing image quality, spatial resolution, operational requirements, and airspace restrictions. Therefore, all flight missions in this study were planned and conducted at a height of 80 m.

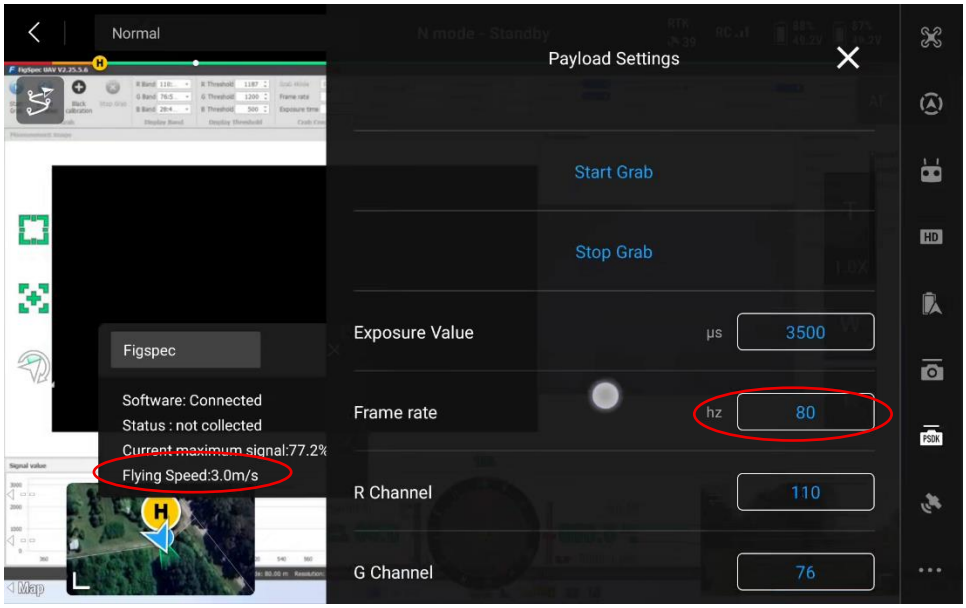


2.7. Set Frame Rate

The manufacturer recommends a frame rate of 50–100 Hz. In this study, the frame rate was set to 80 Hz. Once the Flying Height (m), Frame Rate (Hz) and Spatial Merge (1/2/4) are set, the software automatically calculates the target GSD (cm/pixel) and the recommended Flying Speed (m/s), which is displayed in the real-time data window. This flying height should then be entered into the UAV flight mission.

$$GSD \approx 0.023 \times FlyingHeight \times SpacialMerge$$

$$FlyingSpeed \approx GSD \times FrameRate \div 100$$

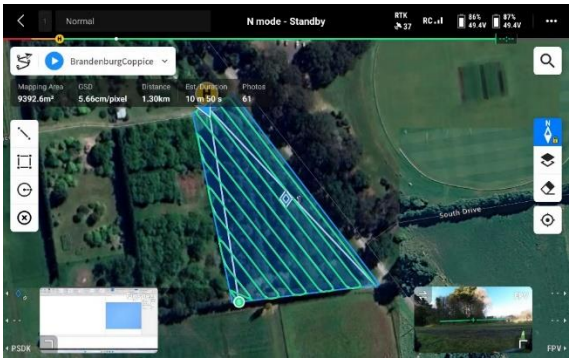


2.8. Set Flight Route

On the flight mission planning page, first set the Route Altitude to match the previously configured Flying Height value, and ensure that the Speed is consistent with the recommended value calculated by the software. Also GSD can be displayed.



Regarding flight route planning, FigSpec recommends planning the flight over a rectangular mission area, with the long side of the rectangle aligned with the flight direction. If the actual study area is irregular, recommended to create a larger rectangle that fully covers the area and then need to crop the imagery afterwards. The lengths of adjacent flight lines should be kept as consistent as possible.

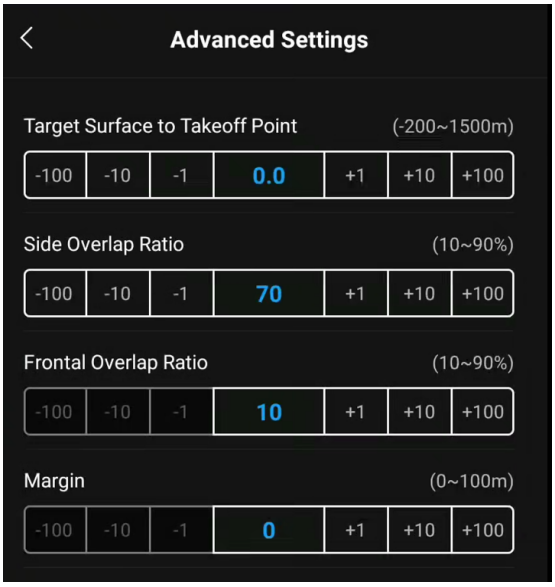


(a) Significant differences in length between adjacent flight lines.



(b) Consistent lengths of adjacent flight lines. (Recommended)

The recommended Side Overlap Ratio and Frontal Overlap Ratio settings are 70% and 10%, respectively.

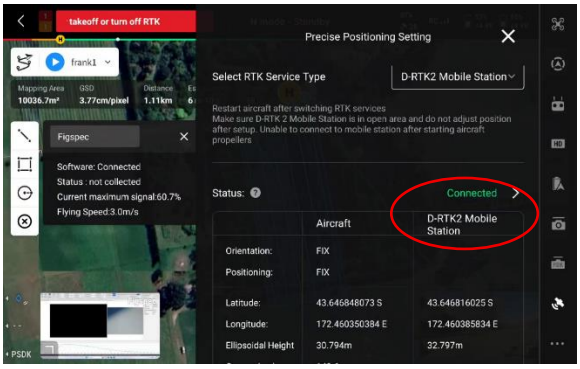


2.9. Pre-Flight Check

After completing the flight route planning, check the weather conditions to ensure they are suitable for the flight mission. It is recommended to avoid flying in Beaufort Force 4 winds or stronger, as even if the drone can operate safely, wind-induced tree movement may cause image blurring and reduce image sharpness and overall data quality. Before takeoff, a thorough pre-flight inspection should also be conducted.

A. Check drone status before takeoff
B. Check the drone wing fastener
C. Check the drone tripod fastener
D. Check the drone battery retainer
E. Check for obstacles directly above the drone
F. Check for obstacles within 3 meters of the drone
G. Open RTK in remote control
H. Check the gimbal mounting bracket's vibration-damping balls.



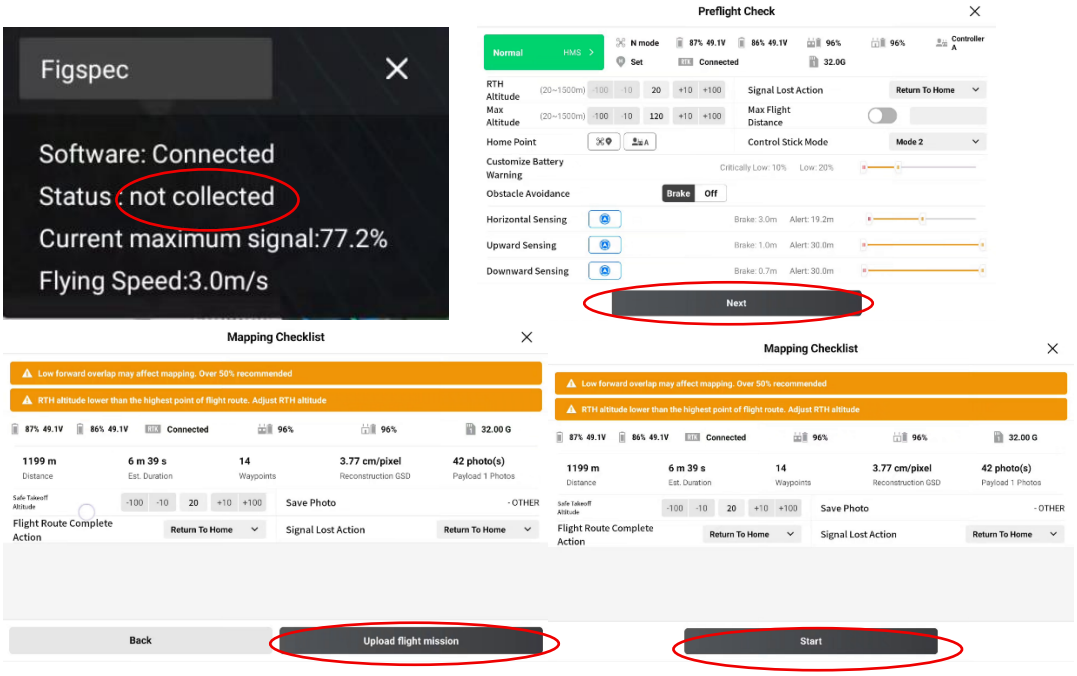




(a) RTK connected (b) Damaged vibration-damping ball with white fluid leakage.

3. Start Flight Mission

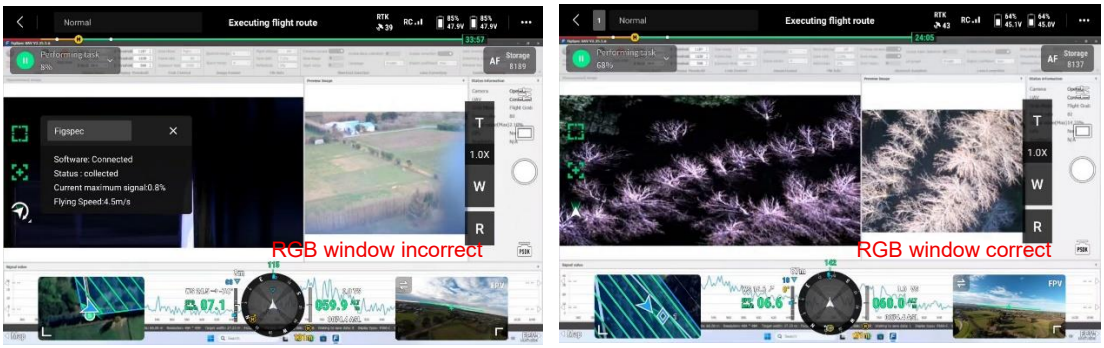
Before takeoff, confirm once again that the white board is positioned within the planned flight path and that the Status in the real-time data window displays “Not Collected”. Then, load the flight mission by following the sequence shown in the figure below.



During takeoff, the drone may occasionally issue a forward obstacle warning, which may be caused by nearby trees obstructing the obstacle detection sensors. If this occurs, the drone can be manually flown to an altitude above the surrounding treetops, provided it is safe to do so. The planned flight mission can then be loaded.

Once the drone reaches the starting point of the planned survey area, the remote controller will provide the voice prompts “Arrived Start Point” and “Starting Test”. Wait approximately two seconds before clicking Start Grab to begin data acquisition. The figure below illustrates an incorrect camera orientation. If the camera is pointing forward and

downward rather than vertically downward, click Stop Grab, then click Start Grab again to restart data acquisition. Confirm that the camera is correctly oriented before continuing.



(a) pointing forward and downward (Incorrect) (b) pointing vertically downward (correct)

When the drone completes the planned flight mission, the remote controller will announce “Mission Completed”. Click Stop Grab to terminate data acquisition.

After the drone has landed and the propellers have completely stopped rotating, shut down the equipment in the following sequence:

1. Turn off the camera power.
2. Turn off the drone power.
3. Turn off the remote controller.
4. Turn off the GPS/RTK base station power.

If you encounter an emergency, click the "H" button in the lower left corner, and then the drone cancels the automatic return and hovers in the air. After the emergency is removed, you can manually control the drone back, or long press the "H" button in the lower left corner, and the drone will continue to return automatically.



4. Data Copying and Preprocessing

4.1. Data Copying

1. Attach an external power to the camera.



2. Attach an external screen to the camera.

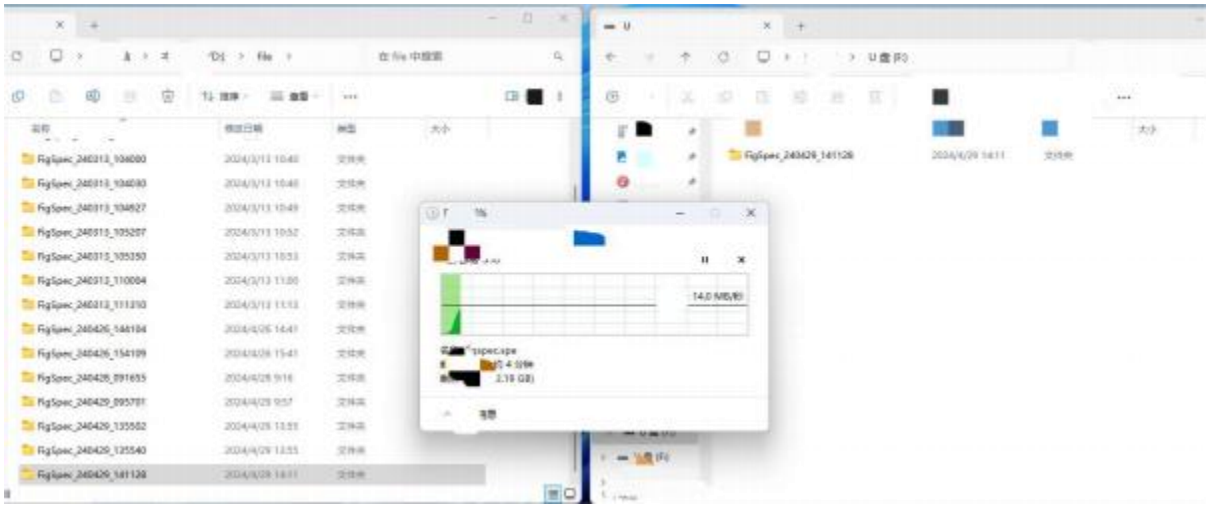


3. External USB splitter, connect the mouse, keyboard, mobile hard disk on the splitter.





4. Open the folder where the data is saved and copy the collected data to the mobile hard drive.



5. Then the camera is turned off and the data is processed on their own computer. The whole collection of data is over.

6. After copying and backing up the collected data, ensure that any unnecessary files are removed to free up storage space. Use the built-in Windows Disk Cleanup utility, located at C:\Windows\System32\cleanmgr.exe, to clean up the storage drive and ensure that sufficient space is available for data acquisition during the next flight mission.

7. Do not forget to regularly recharge the remote controller's internal battery via the USB Type-C port. The internal battery is non-removable and must be charged directly through this port.

4.2. Data Preprocessing

In this study, two official FigSpec software packages, FigSpec Merge 2.0 and FigSpec Studio 2.0, were primarily used for data merging and white reference board calibration, respectively. The following sections will also introduce the use of ENVI 5.6 to extract the spectral bands required for subsequent analysis.

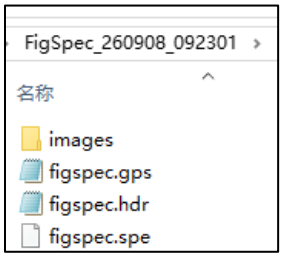


Figure4.1. Raw Data Files Collected During Acquisition

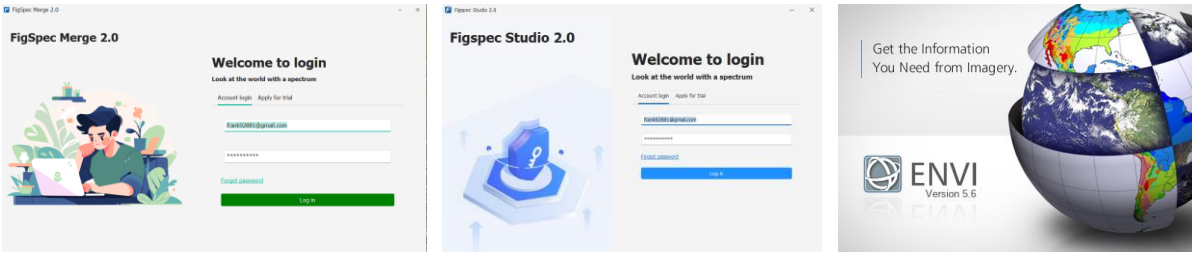


Figure4.2. Preprocessing Softwares

The software licences are not tied to a specific computer, and there are no restrictions on the number of simultaneous logins. Therefore, the software can be installed and used on multiple computers without additional activation requirements.

The login credentials for the two official FigSpec software packages are as follows:

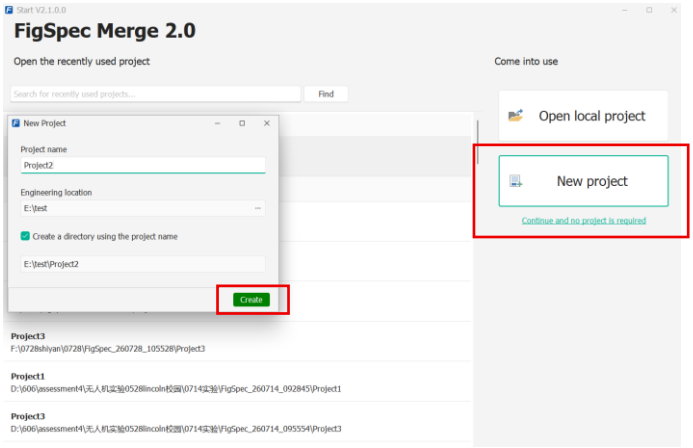
- **Account: frank92881@gmail.com**
- **Password: frank92881**
- Licence Expiration Date: 1 January 2050

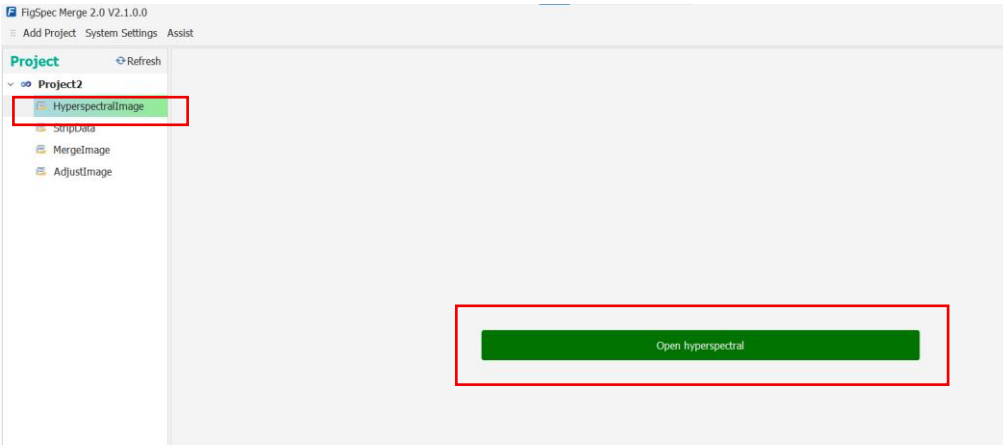
Important Note on Disk Space Requirements:

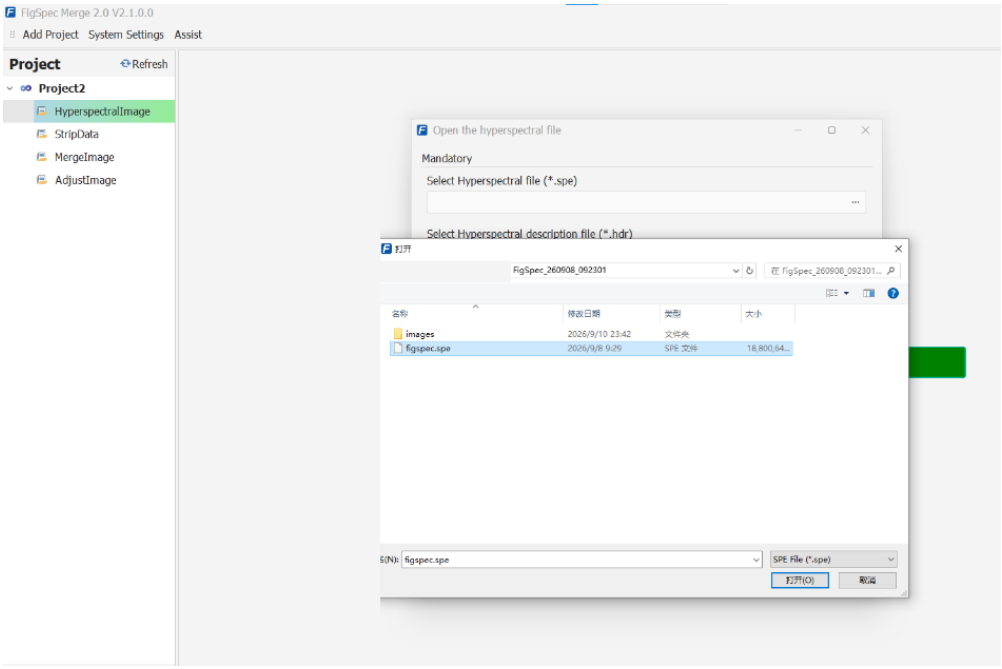
The raw spectral data files (.spe files) collected during flight missions are typically very large, often reaching tens of gigabytes (GB). When a new project is created in official softwares and the raw spectral files are imported, the software automatically copies the original spectral files into the new project directory. This mechanism helps protect the original data from accidental modification or overwriting. However, it also means that disk space usage can approximately double after importing the data. Furthermore, each time a new project is created and the same raw spectral files are imported, the software generates another copy of the source files within the corresponding project directory, resulting in additional storage consumption. Therefore, always ensure that the destination drive and project directory have sufficient available disk space before creating a new project or importing raw spectral data.

4.2.1 FigSpec Merge 2.0

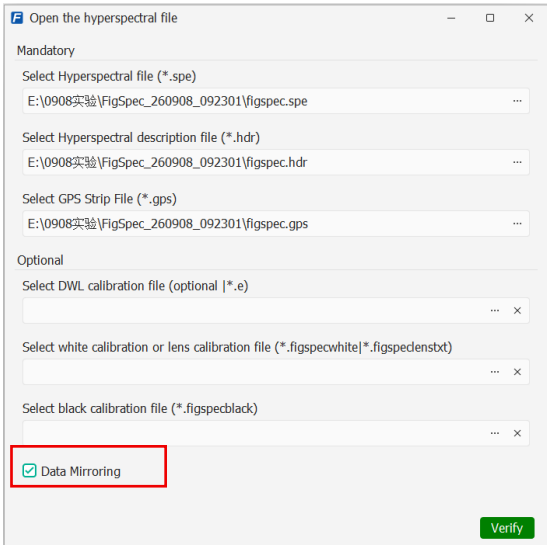
Open FigSpec Merge 2.0 and create a new Project 2. Click **Open Hyperspectral**, then select the collected raw spectral data file (.spe).



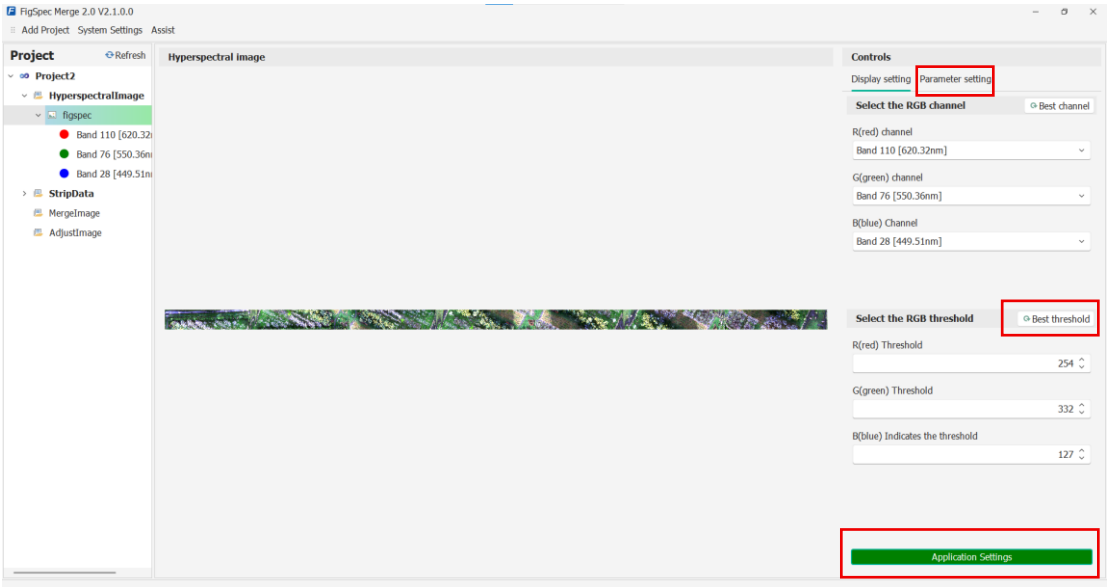




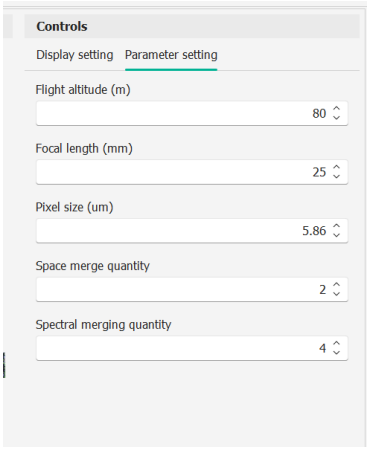
After importing the **.spe file**, the corresponding **.hdr** and **.gps** files will be automatically matched. Make sure to select the **Data Mirror option**, as this is essential for correct image processing. Then, click **Verify** to proceed.



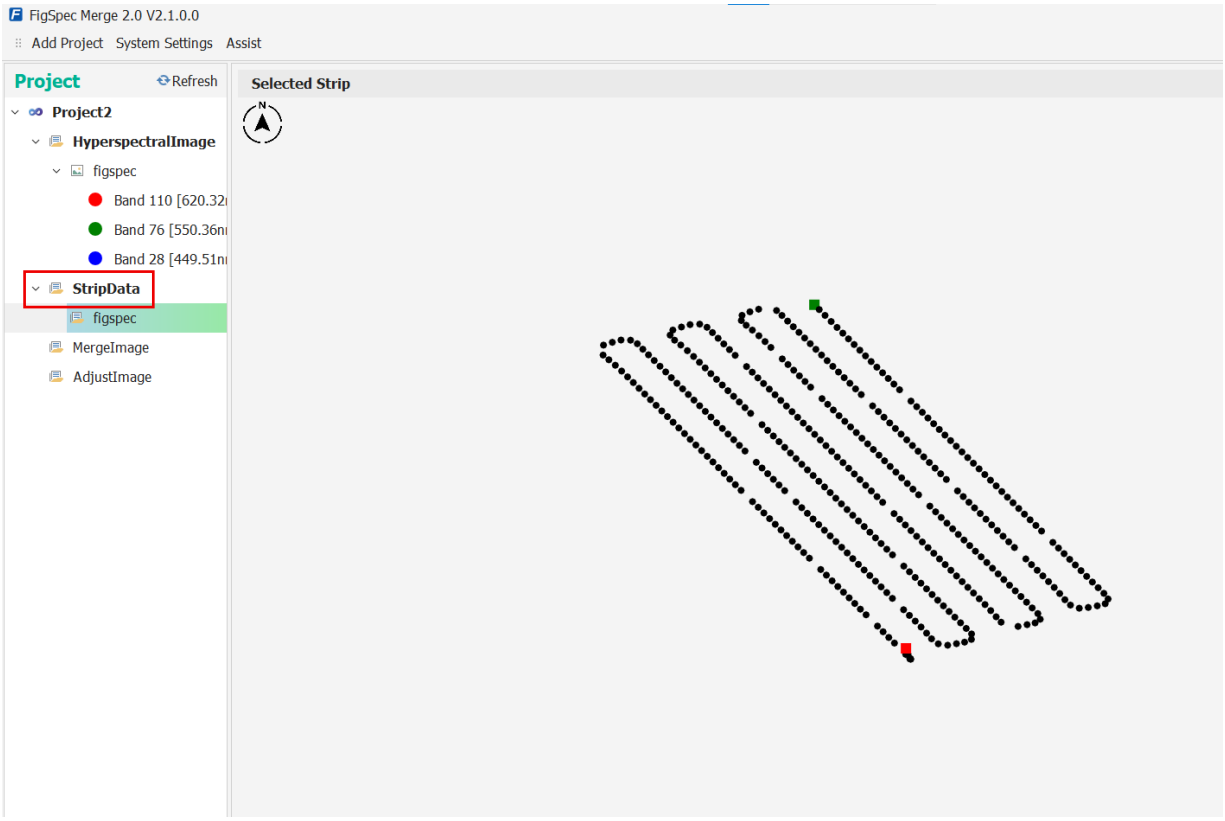
After importing the data, the software will display the entire flight dataset as a single continuous flight strip, including all flight lines and turning sections. At this stage, all flight lines are combined and displayed as one continuous strip rather than as separate strips. Click **Best Threshold** at the bottom of the interface, then click **Application Setting**. The software will automatically adjust the image brightness to improve its visual appearance.



Click **Parameter Settings** to display the flight configuration parameters associated with the imported dataset.



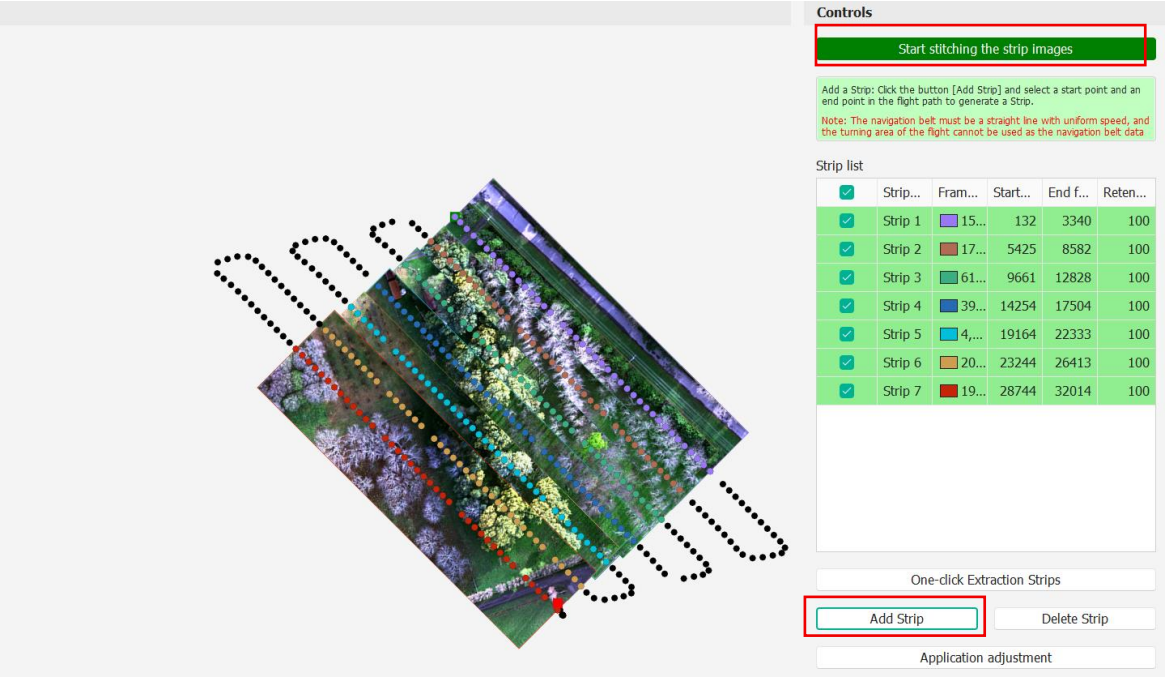
Click **StripData** on the left panel to display the flight routes.



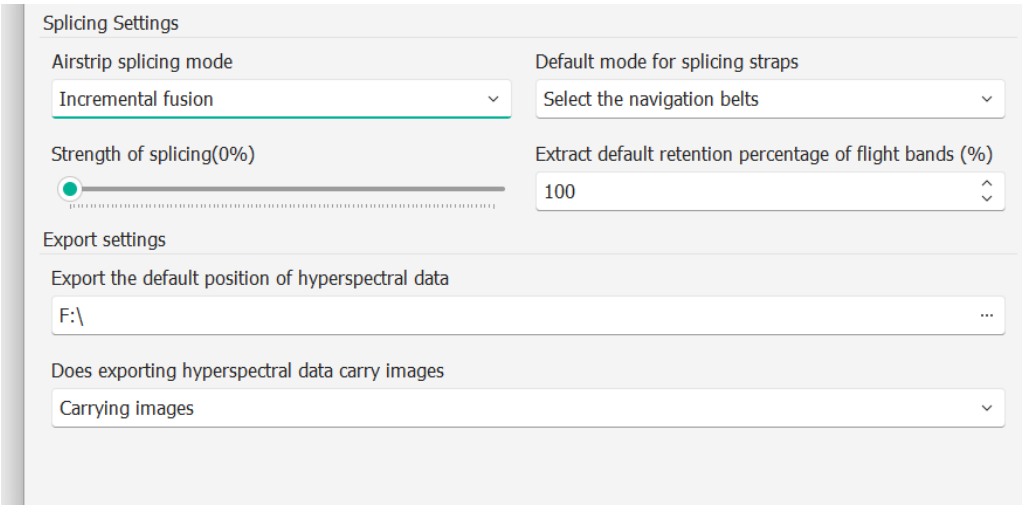
Click **Add Strip**, then use the mouse to select the start and end points of each individual flight strip to extract it from the continuous flight data. Ensure that the selected section is straight and does not include any turning sections.

When extracting adjacent flight strips, try to keep their lengths as consistent as possible, as this helps improve the accuracy of subsequent image stitching and mosaicking.

As shown in the figure, adjacent flight strips contain overlapping areas, which are used for image alignment and stitching during the mosaicking process.



In **System Settings**, the following parameter configurations are recommended.



Click **Start Stitching the Strip Images** to begin mosaicking the adjacent flight strips. The figure below shows the mosaicked image generated from seven adjacent flight strips.

As shown in the figure below, slight brightness differences can be observed along the seams between adjacent strips after stitching. These inconsistencies are primarily associated with limitations in the current software and cannot yet be completely eliminated.

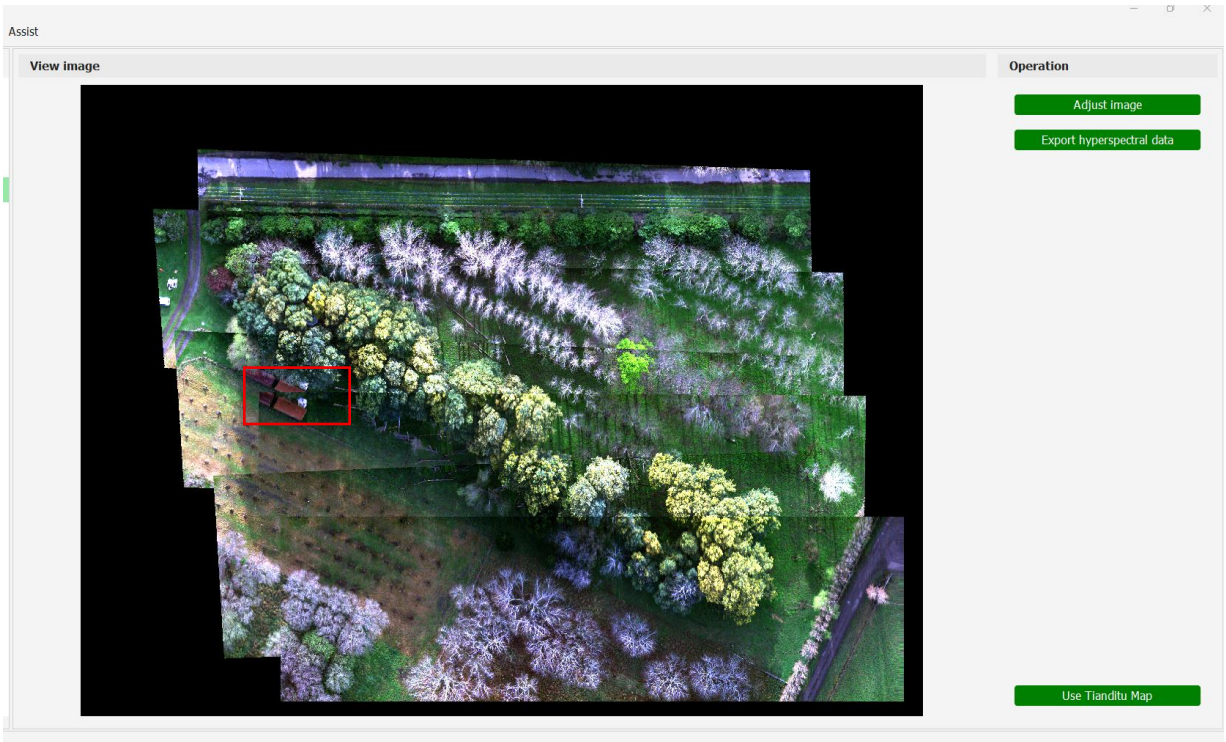
In addition, noticeable spatial misalignment can occur along some strip boundaries, particularly in areas with substantial variations in surface elevation and canopy height. For example, in the figure below, the area surrounding the shipping containers consists mainly of bare ground, while the adjacent tree canopy reaches approximately 30 m in height. This significant height difference introduces height-induced parallax, resulting in inter-strip

misalignment during the mosaicking process. Such geometric distortions are particularly problematic in heterogeneous forest environments where tall vegetation and exposed ground coexist within the same flight strip.

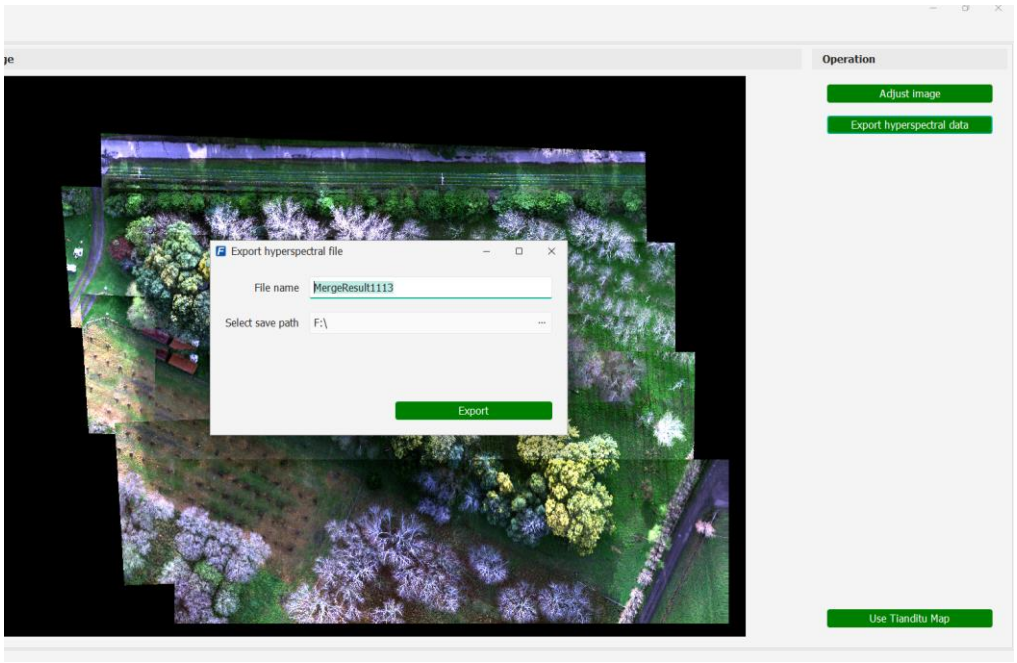
The manufacturer's technical support team has also confirmed that FigSpec Merge 2.0 still contains some software limitations and is undergoing continuous development and improvement. At present, image stitching may fail in complex environments with substantial variations in canopy height and surface features. In contrast, relatively homogeneous landscapes, such as flat agricultural fields, are generally easier to mosaic successfully. However, the study area in this project contains a heterogeneous mixture of tree canopies, vegetation, and exposed ground, making accurate image stitching considerably more challenging.

According to the manufacturer, future updates to FigSpec Merge may introduce improved image-matching and geometric correction algorithms using Google satellite imagery as a geographic reference. Such improvements could substantially enhance the accuracy of image alignment and mosaicking, helping to minimise inter-strip misalignment caused by height-induced parallax, particularly in forested areas with large differences between canopy and ground elevations.

However, it should be noted that satellite-image-based registration alone may not completely eliminate height-induced parallax. More accurate geometric correction may require incorporating terrain and canopy height information, such as a Digital Elevation Model (DEM) or Digital Surface Model (DSM), into the orthorectification process.



Click **Export Hyperspectral Data** to export the stitched hyperspectral dataset. The exported file is named MergeResult1113.spe.

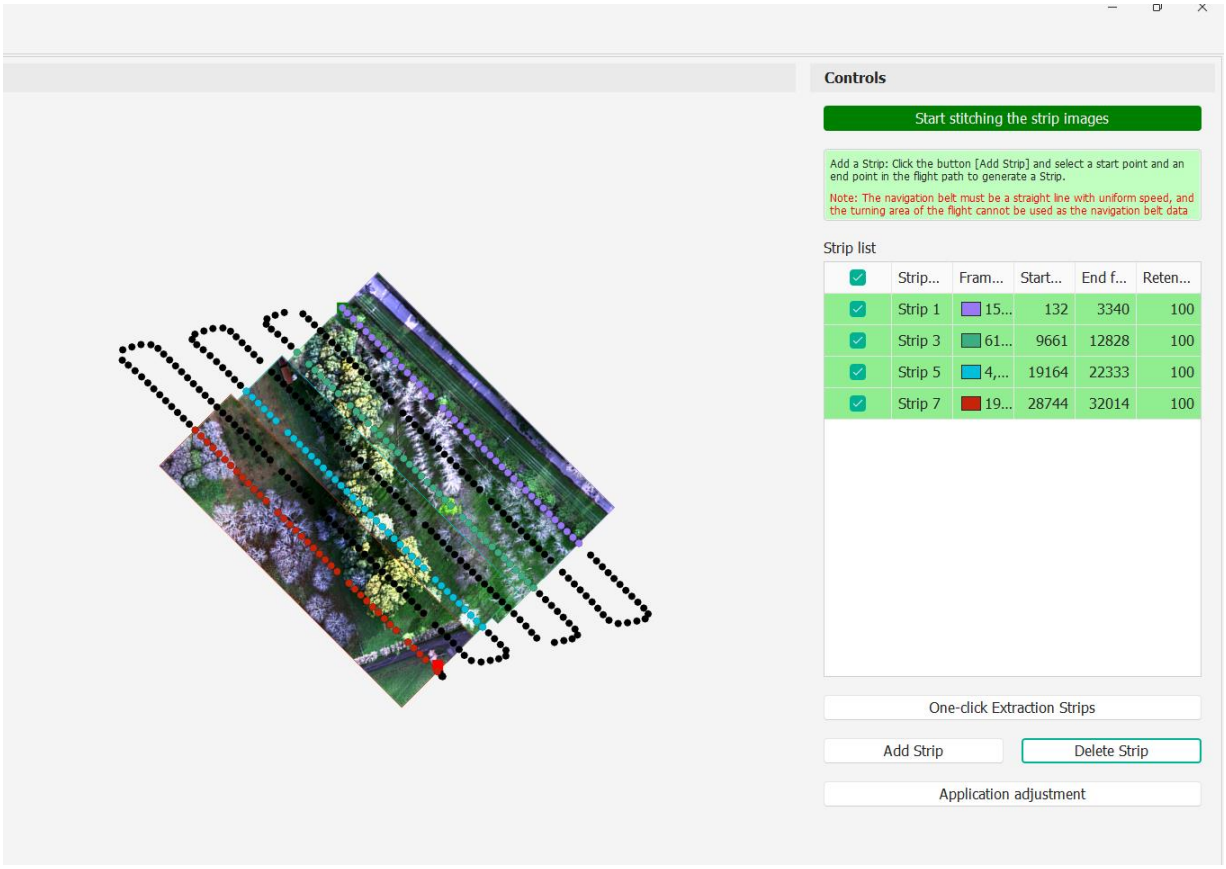


When selecting flight strips for mosaicking, an alternative approach is to exclude some adjacent strips and adopt an alternating-strip mosaicking strategy.

Because adjacent flight strips have a relatively high overlap ratio, stitching all strips together (e.g., Strip 1 + 2 + 3 + 4 + 5 + 6 + 7) may introduce excessive redundant information and potentially increase the complexity of image matching, resulting in inter-strip misalignment. As illustrated in the figure below, selecting alternating flight strips (e.g., Strip 1 + 3 + 5 + 7) may produce better mosaicking results by reducing redundant overlap and potential alignment errors.

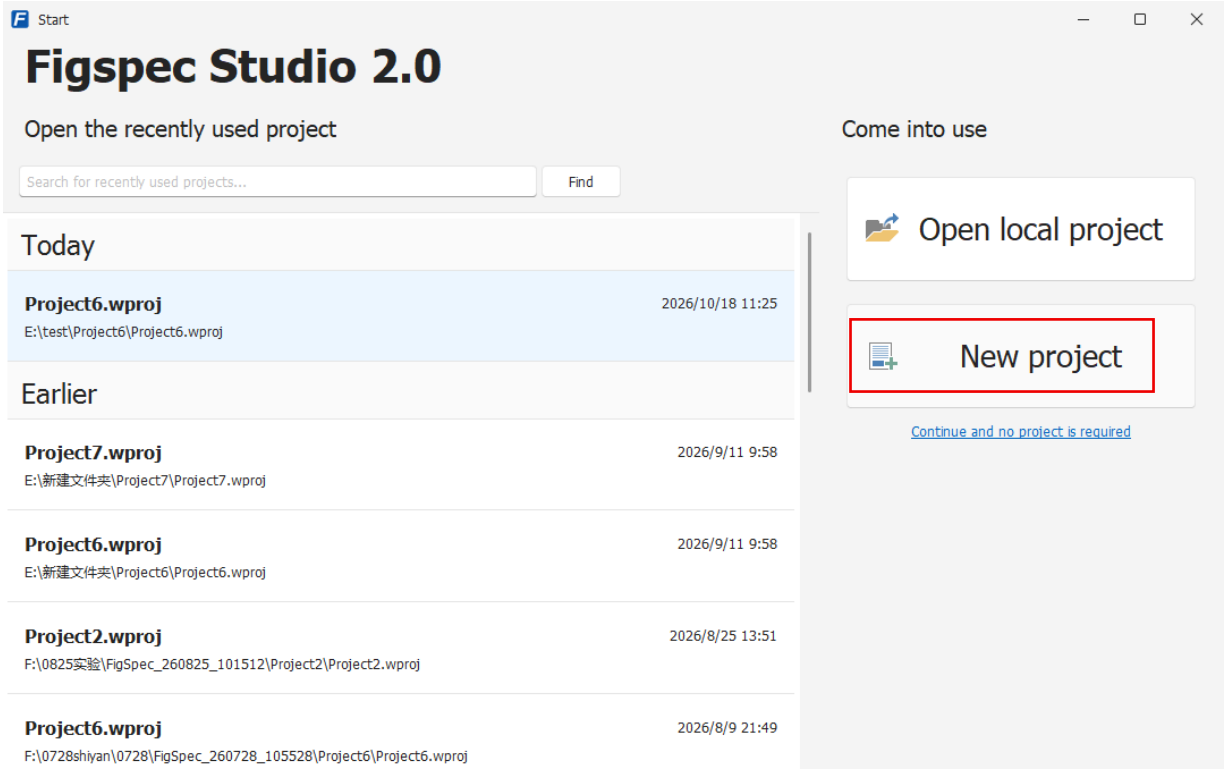
However, this approach is still experimental and requires further investigation and validation to determine its effectiveness and reliability under different flight and terrain conditions.

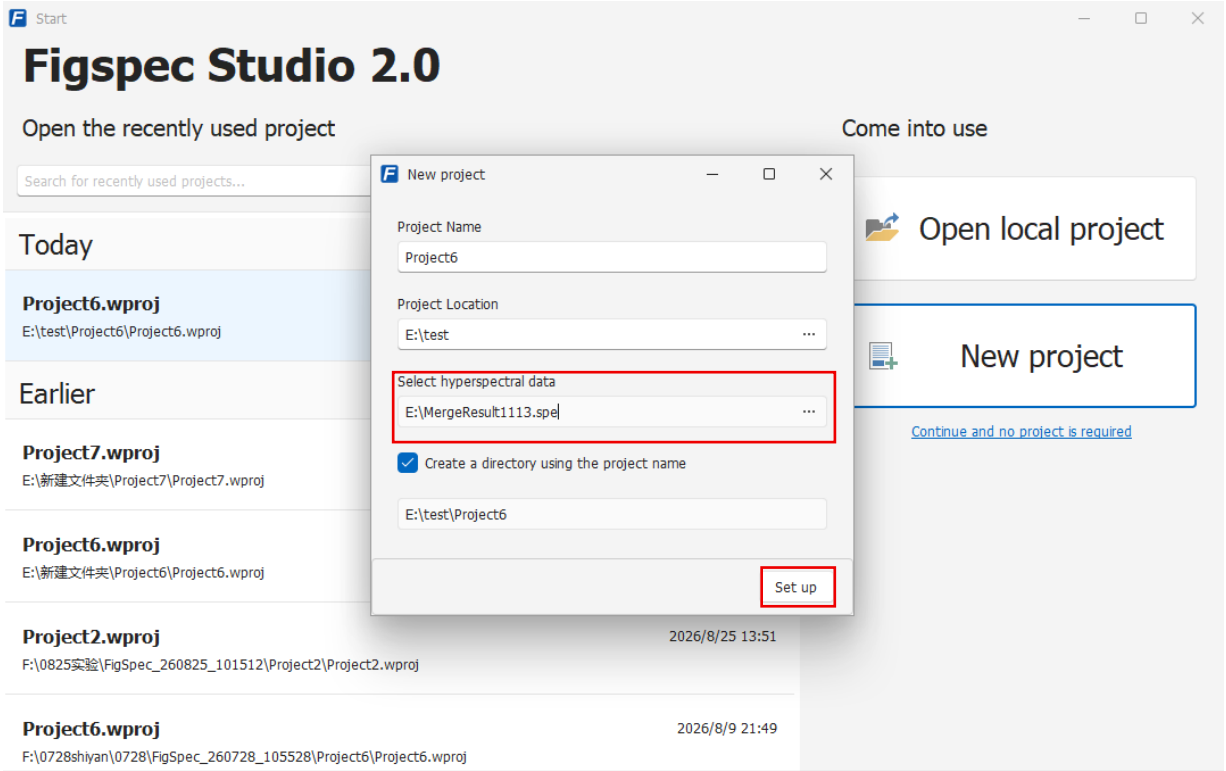
After completing the mosaicking process, close FigSpec Merge 2.0.


Controls
Start stitching the strip images
Add a Strip: Click the button [Add Strip] and select a start point and an end point in the flight path to generate a Strip.
Note: The navigation belt must be a straight line with uniform speed, and the turning area of the flight cannot be used as the navigation belt data
Strip list
Strip... Fram... Start... End f... Reten...
Strip 1 15... 132 3340 100
Strip 3 61... 9661 12828 100
Strip 5 4,... 19164 22333 100
Strip 7 19... 28744 32014 100
One-click Extraction Strips
Add Strip
Delete Strip
Application adjustment

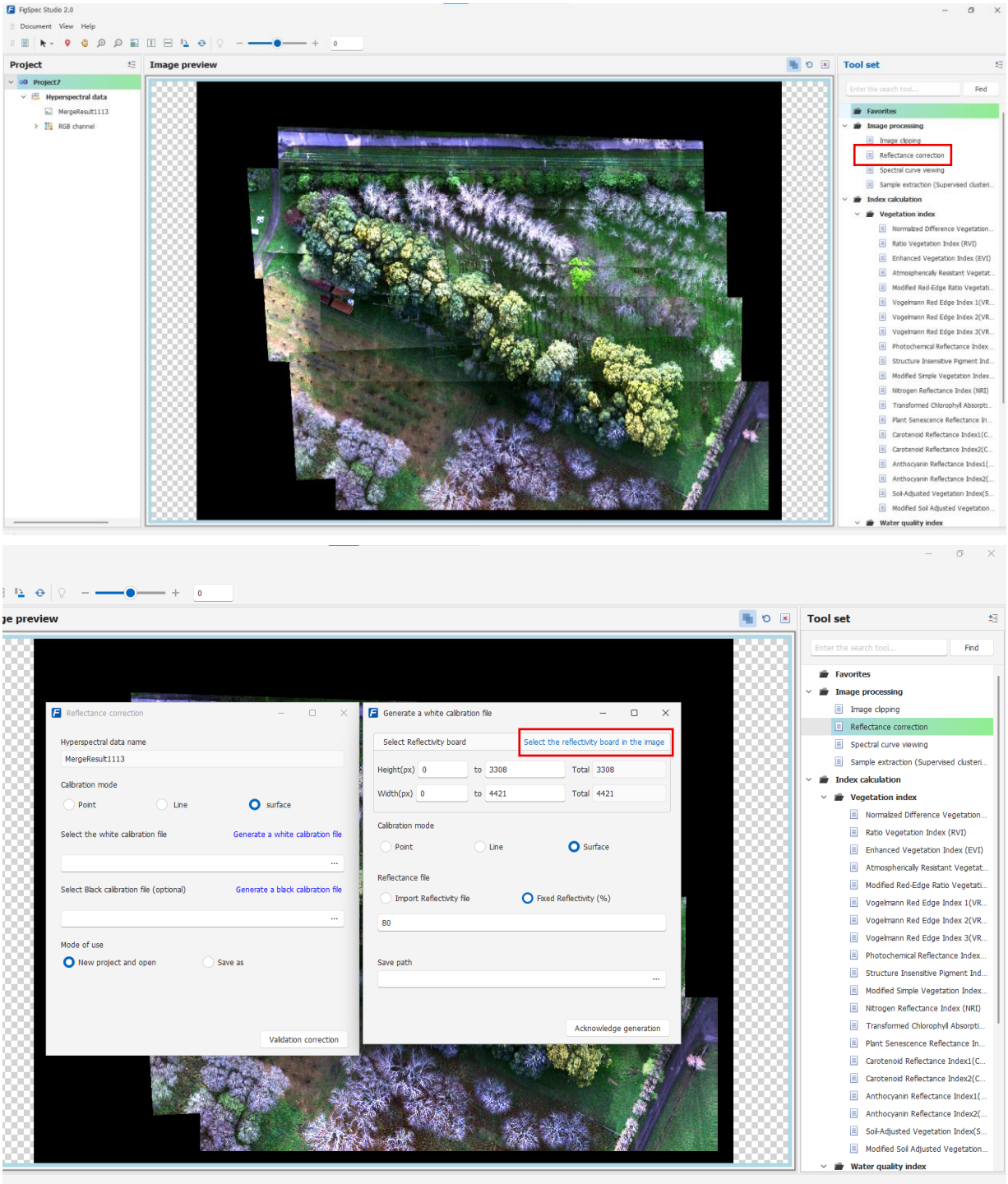
4.2.2 FigSpec Studio 2.0

Open FigSpec Studio 2.0 and create a new project 6. Then, select and import the previously mosaicked hyperspectral data file, MergeResult1113.spe.



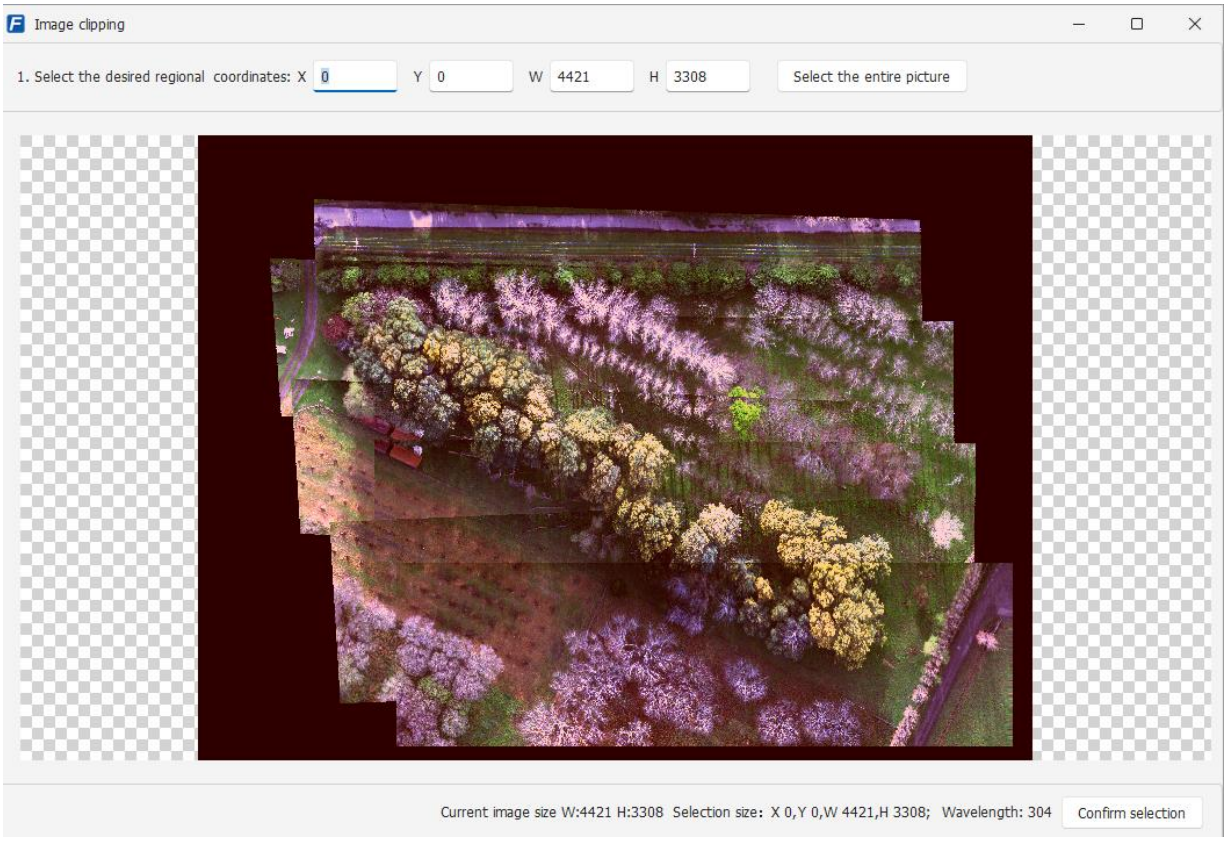


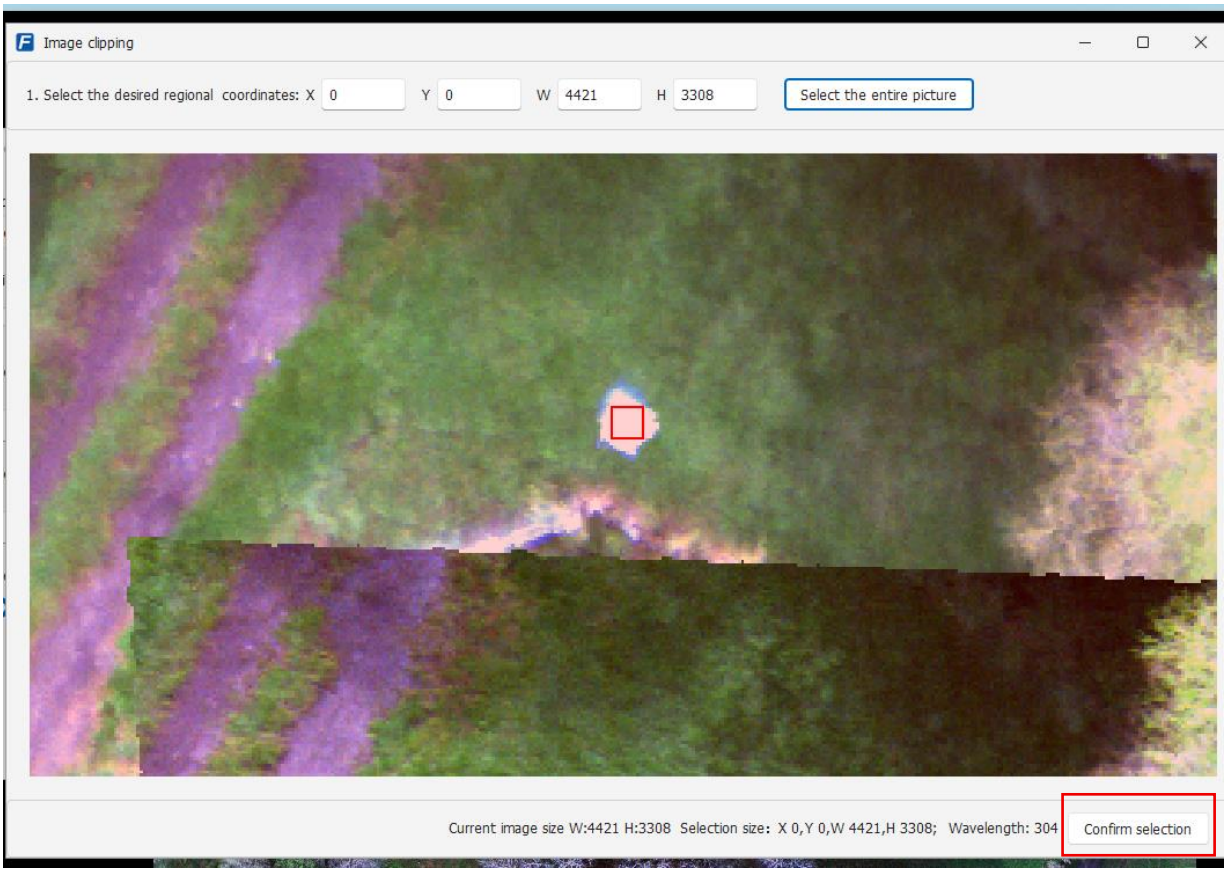
Once the mosaicked image is displayed, click **Reflectance Correction** on the right-hand panel and configure the settings as shown below.



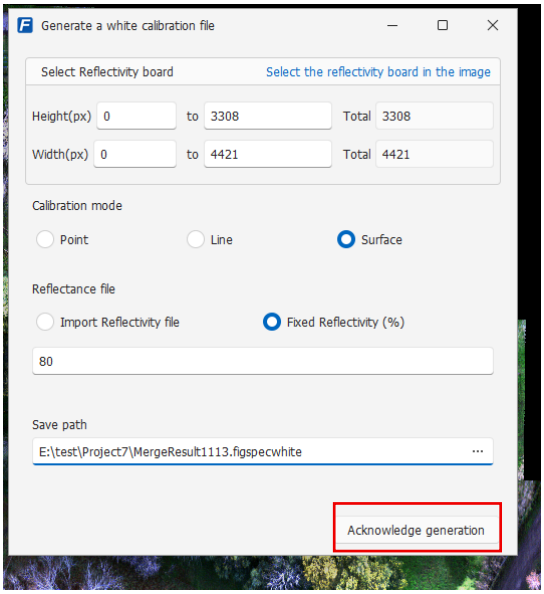
Click **Select the reflectivity board in the image**, then use the mouse scroll wheel to zoom in on the image. Locate and select the white board, then click Confirm Selection to complete the

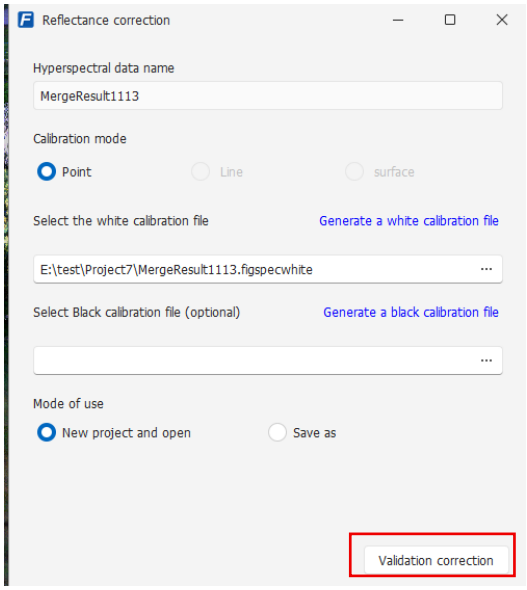
selection process.



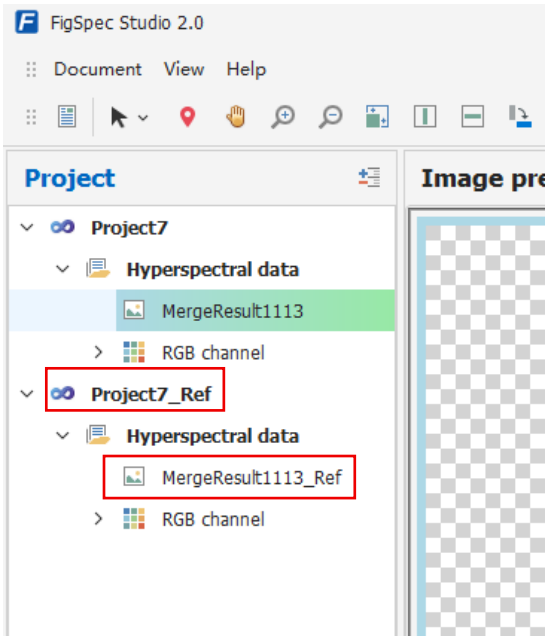


Click **Acknowledge generation**, Then **Click Validation Correction**. Wait until the progress bar reaches 100% and the process is complete.

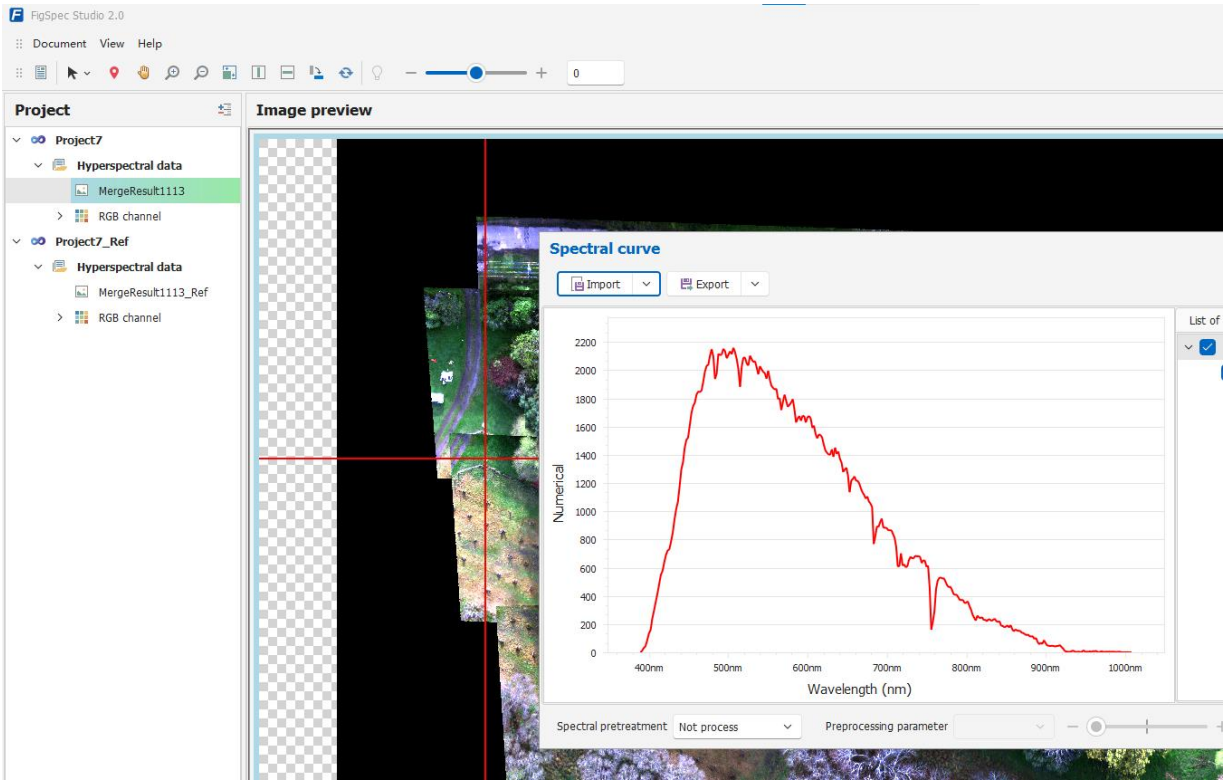


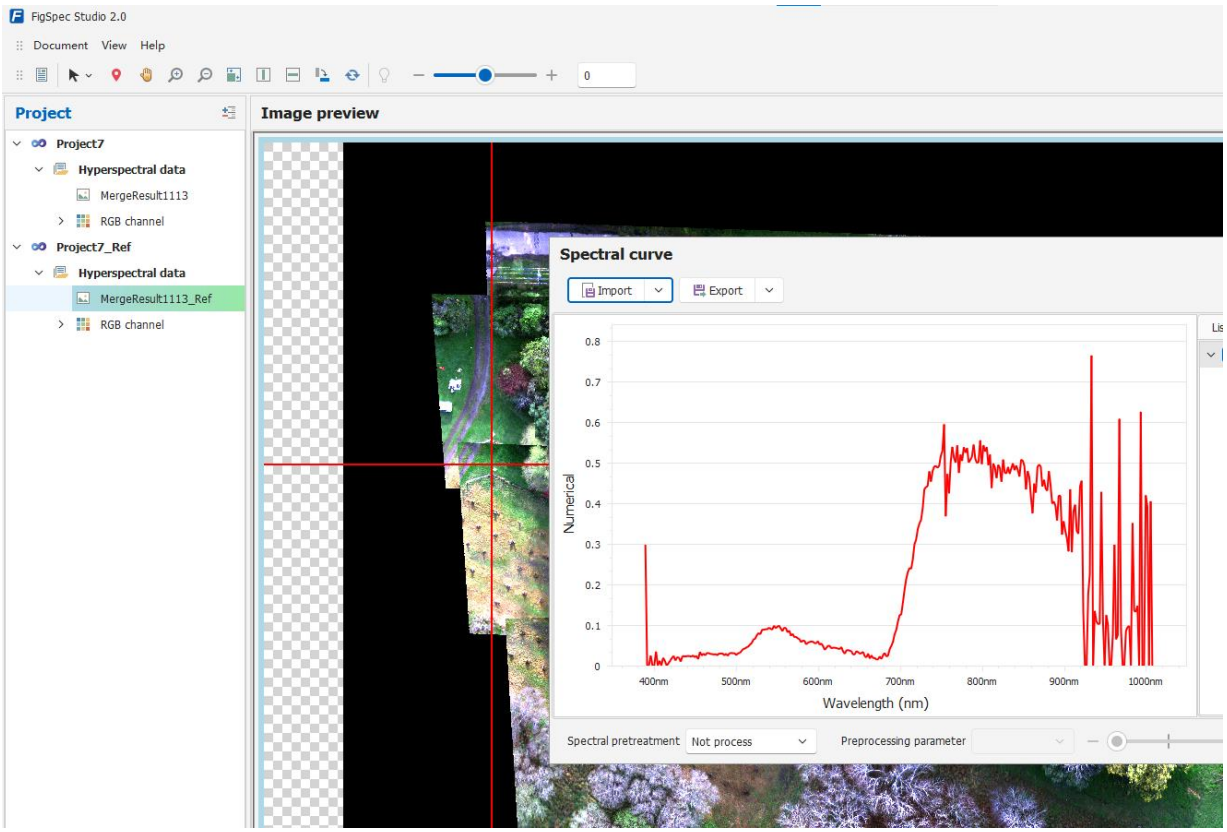


A new project named Project7_Ref will appear in the left-hand panel. The newly generated reflectance-corrected file, MergeResult1333_Ref, will be displayed beneath it.

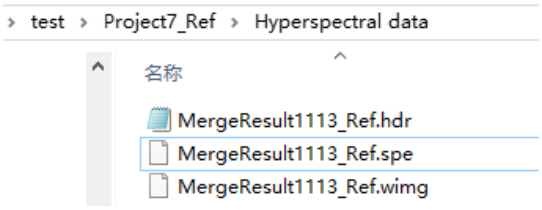


Compare the original Merge file with the newly generated Merge_Ref file. The comparison shows that the spectral values on the Y-axis have been normalised and calibrated through the reflectance calibration process.

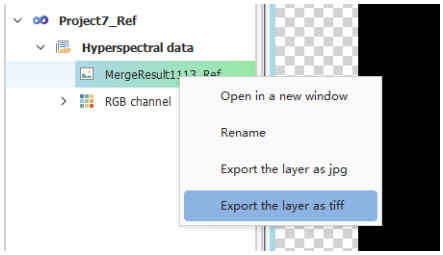

FigSpec Studio 2.0
Document View Help
Project
Project7
Hyperspectral data
MergeResult1113
RGB channel
Project7_Ref
Hyperspectral data
MergeResult1113_Ref
RGB channel
Image preview
Spectral curve
Import
Export
Numerical
Wavelength (nm)
Spectral pretreatment
Not process
Preprocessing parameter


FigSpec Studio 2.0
Document View Help
Project
Project7
Hyperspectral data
MergeResult1113
RGB channel
Project7_Ref
Hyperspectral data
MergeResult1113_Ref
RGB channel
Image preview
Spectral curve
Import
Export
Numerical
Wavelength (nm)
Spectral pretreatment
Not process
Preprocessing parameter

The software will automatically generate the reflectance-calibrated file, MergeResult1113_Ref.spe, in the corresponding output directory.

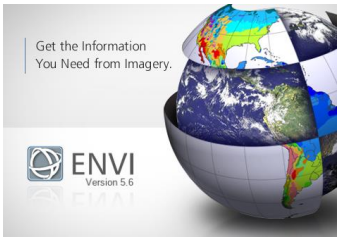


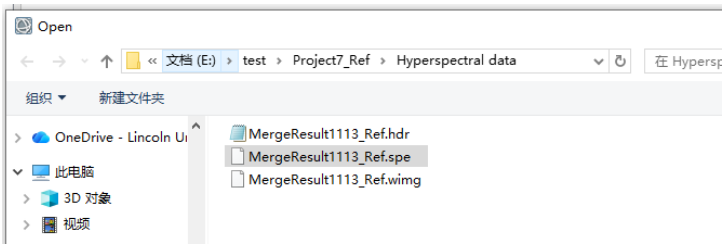
Important Note: Do not use the **Export TIFF** function at this stage, as it only exports a three-band RGB TIFF image, rather than a multi-band hyperspectral TIFF file containing the full spectral information.



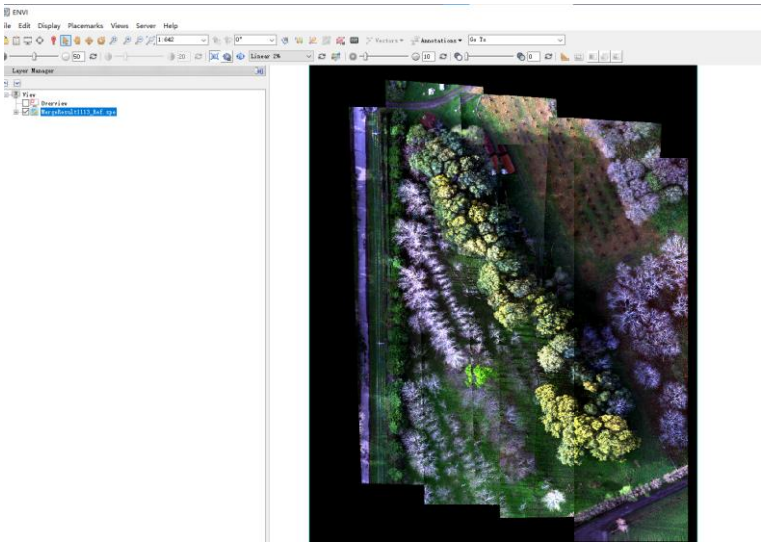
4.2.3 ENVI 5.6

Open ENVI 5.6, then load the newly generated reflectance-calibrated hyperspectral data file, MergeResult1113_Ref.spe.



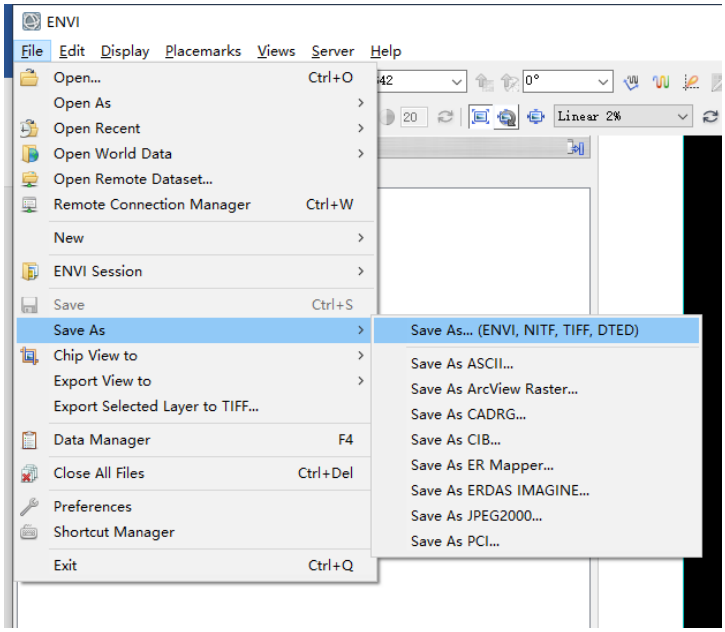


The mosaicked and reflectance-calibrated hyperspectral image will now be displayed in ENVI 5.6.



Click **Save As** and select the required spectral bands for export. Table 1 provides a reference list of commonly used spectral bands and their corresponding wavelengths. To select multiple bands, hold down the Ctrl key and left-click the desired bands.

In this study, the spectral bands were selected and exported according to the specifications listed in Table 1.



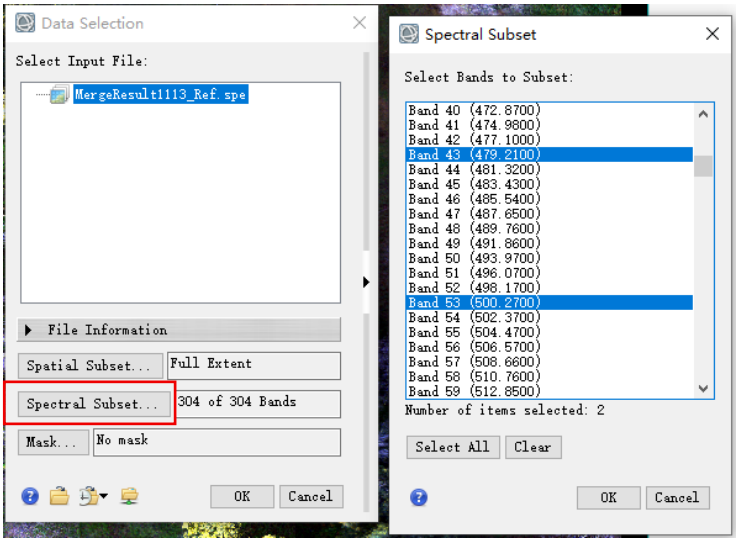
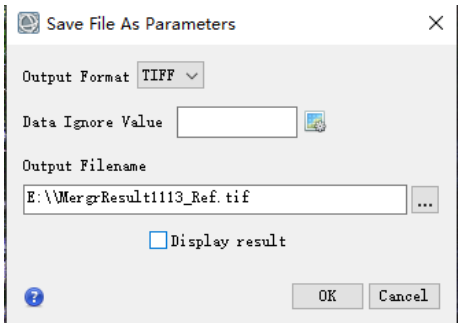


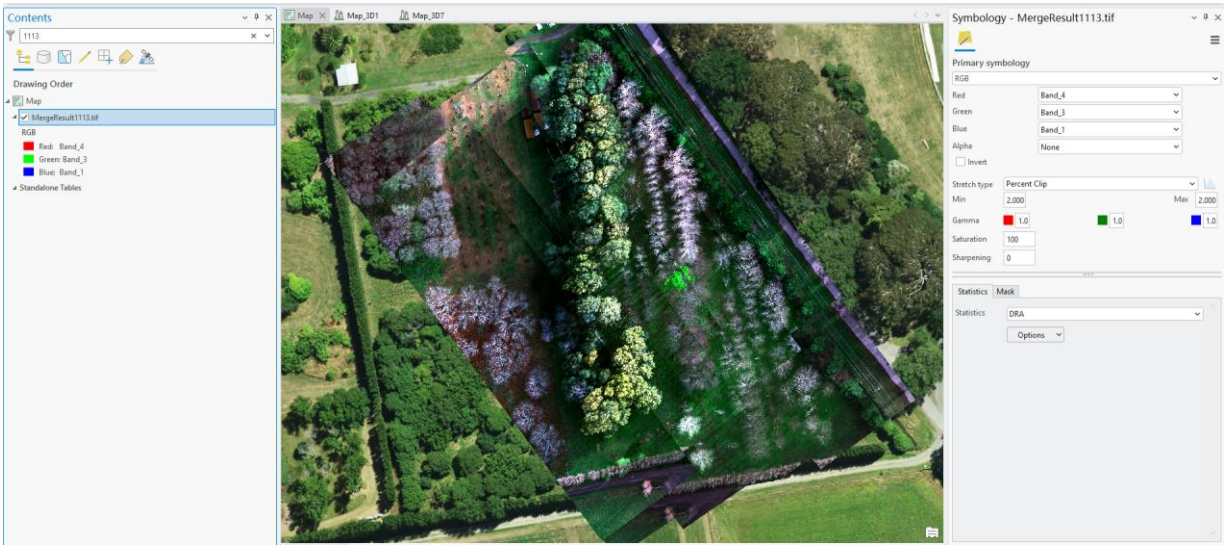
Table 1. Commonly Used Hyperspectral Bands and Their Corresponding Wavelengths for Spectral Analysis

Wavelength	Main Application	Band in ENVI	Band in Arcgis
480 nm	Blue light; RGB, EVI	43	1
500 nm	PSRI; leaf senescence analysis	53	2
560 nm	Green light; RGB, GNDVI	82	3
670 nm	Red absorption band; NDVI, EVI	135	4
680 nm	PSRI	140	5
705 nm	Red-edge band; NDRE	153	6
740 nm	Post-red-edge region; canopy changes	170	7
750 nm	PSRI reference band	175	8
800 nm	Near-infrared (NIR); NDVI, GNDVI, NDRE	200	9
900 nm	WBI reference band	250	10
970 nm	Water absorption band; WBI	285	11

After selecting the required spectral bands, configure the export settings as shown in the figure below, then click **OK** to generate a multi-band TIFF (.tif) file containing the selected spectral bands.

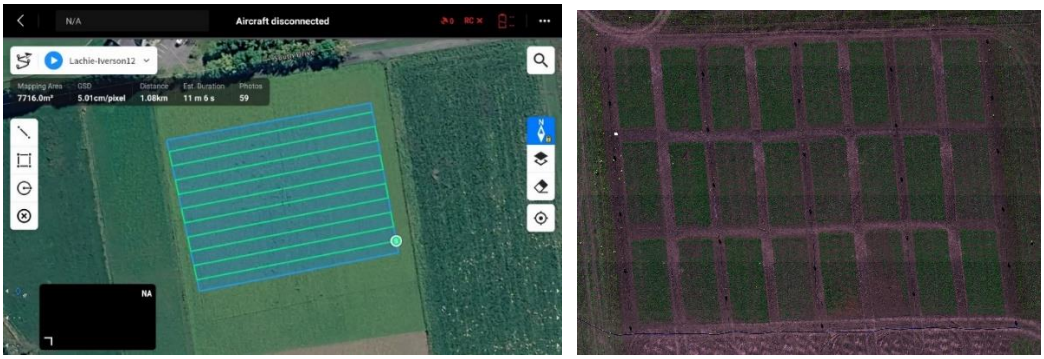


Import the generated multi-band TIFF (.tif) file into ArcGIS Pro, as shown in the figure below. **Important Note:** When configuring the RGB display, ensure that the Red, Green, and Blue channels are assigned to Band 4, Band 3, and Band 1, respectively, according to the band mapping provided in Table 1.



4.2.4 Exploration

The current software produces significantly better mosaicking results in relatively flat and homogeneous areas than in areas with substantial variations in terrain elevation and canopy height, as illustrated in the figure below.



(a) flat and homogeneous areas





(b) variations in terrain elevation and canopy height

Exploration of an Alternative Mosaicking Method

An alternative mosaicking approach is to export each flight strip individually and import the resulting files into ArcGIS Pro. Each strip can then be manually georeferenced using Google satellite imagery as a reference.

Once all individual flight strips have been georeferenced and reflectance-calibrated using the white board, the Mosaic To New Raster tool in ArcGIS Pro can be used to merge the adjacent strips into a single mosaicked image.

This approach may provide better spatial alignment and mosaicking accuracy than the automatic stitching process in FigSpec Merge 2.0, particularly in areas with substantial variations in canopy height and terrain elevation. However, it is considerably more time-consuming, as each flight strip requires individual georeferencing and processing.

Note: Although this method may improve inter-strip alignment, manual georeferencing against satellite imagery does not fully eliminate height-induced parallax. Additional geometric correction using elevation data may still be required to achieve accurate orthorectification in complex forest environments.

5. Acknowledgements

I would like to express my sincere gratitude to Helen and Tim for her guidance and support throughout this project.

Special thanks are extended to Lekshmi and Abin for their valuable technical advice and assistance with hyperspectral imaging and data processing.

I would also like to thank Lanchie, James, Donald, and Stuart for their assistance, technical support, and contributions during the project.

My appreciation also goes to the FigSpec Technical Support Team for their assistance in troubleshooting software issues, explaining technical limitations, and discussing potential improvements to the image mosaicking process.

This guide was independently compiled based on my practical experience with the FigSpec FS60-C hyperspectral camera, including field operations, flight testing, data processing, troubleshooting, and the exploration of alternative image mosaicking methods. The procedures, observations, and recommendations presented here primarily reflect the experience and lessons learned during this project, with contributions and advice from the individuals acknowledged above.